

Operationalizing Screening Off for Rainforest Admission

Abstract

Scale-relative structural realism allows that real patterns may occupy more than one level of ontology, but the admittance criteria for those levels remain underdetermined. Franklin and Robertson (2024) offer the most developed recent proposal: emergence, understood as screening off plus novelty, earns an entity its place in the rainforest. Their criterion is the right one, but its stated test underdetermines it. Conditional independence admits an observational reading, on which micro-detail is redundant across the system’s actual trajectory, and a modal reading, on which that redundancy survives admissible intervention. They coincide in the central cases and diverge where apparent macro-autonomy is borrowed from off-partition structure. A macro-variable driven by a shared upstream process can satisfy their conditions in full, novelty included, while distinct micro-realizations of its macrostate come apart under perturbation. On the modal reading, screening off and transition autonomy are not two conditions but one: to screen off micro-detail from the macro-future under admissible perturbation is to carry autonomous transition structure at that grain. Closure is the admissibility-indexed test that recovers this modal content. A candidate qualifies when, for a fixed regime, horizon, and admissible intervention class, macrostate information suffices for macro-transitions. Strong lumpability is the exact benchmark; non-ideal cases are graded by leakiness, which must itself be indexed to the intervention class rather than to the observed distribution. Interventions serve throughout as diagnostic probes rather than as metaphysical primitives. The contribution is a correction to how an existing filter is read and applied rather than a rival to it.

Keywords: real patterns, rainforest realism, closure, strong lumpability, structural realism, interventionism

1 Introduction: Closure and the Admittance Problem

Lewis observed that we have no principled reason to deny existence to the mereological sum of “the right half of my left shoe plus the Moon plus the sum of all Her Majesty’s ear-rings,” however sensible it is to ignore such things in ordinary thought (Lewis 1986, p. 213). Fair enough. But consider the difference between that composite and a hurricane. Knowing a hurricane’s pressure organization and rotational structure tells you what it will do next without tracking every molecule. The shoe-moon

composite carries no comparable transition structure; predicting its future requires independently tracking each component. The question is not whether both exist. The question is which patterns carry autonomous macro-dynamics and therefore deserve robust commitment.

Scale-relative structural realism says that the world is not exhausted by a single privileged descriptive level. The special sciences succeed by tracking higher-level patterns stable enough to support prediction, explanation, and intervention, and Ladyman and Ross’s rainforest realism gave this thought its most influential metaphysical form: reality is populated by real patterns at many scales, not only by microphysical furniture (Ladyman and Ross 2007). But one question remains insufficiently settled. Which patterns earn admission into that rainforest, and which should be treated more cautiously as useful summaries, temporary proxies, or dynamically idle aggregates?

Franklin and Robertson offer the most developed recent answer (Franklin and Robertson 2024). Emergence earns rainforest admission when the macro-variable renders micro-details conditionally irrelevant (screening off) relative to the explanandum, and the macro-variable participates in a dependency profile that is not type-identical to the realizing micro-dependencies. That is a substantial advance. It captures why obvious gerrymanders and trout-turkey style composites should be excluded, while preserving the thought that higher-level ontology is earned by stable macro-level dependence rather than by intuitive naturalness alone. They explicitly cast the issue as one of the “admittance criteria for the rainforest” (Franklin and Robertson 2024, §1), and they do so in a way meant to remain compatible with theoretical reducibility rather than to revive a strong irreducibility test.

Screening off plus novelty is indispensable, but admits two readings of conditional independence, observational and modal, that existing admittance tests do not discriminate. A candidate macro-variable deserves robust commitment only under the modal reading: it must carry its own transition structure under a fixed regime, horizon, and admissible intervention class. In exact Markov settings, strong lumpability is the benchmark realization of that condition. In non-ideal settings, closure is graded via leakiness and convergent diagnostics under fixed constraints, with explicit downgrade rules when verdicts are fragile. A candidate can satisfy observational screening-off conditions and enter into robust, law-like macrodependencies while still borrowing its apparent autonomy from upstream structure or from hidden micro-identity. Closure supplies the admissibility-indexed selector that forces the disambiguation.

The dialectical structure is cumulative rather than competitive. Dennett’s real patterns identify candidate structures through compression and projectibility (Dennett 1991); Franklin and Robertson supply the admittance filter by requiring screening off plus novelty (Franklin and Robertson 2024); the criterion developed here specifies which reading of that filter earns admission and provides the test that discriminates it. The candidates it withholds commitment from are those that pass the filter observationally while remaining parasitic on off-partition structure, whether hidden micro-identity, a shared upstream cause, or a fast hidden relaxation process. The aim is to sharpen rainforest realism from inside its own commitments rather than to displace them.

The wedge that forces the further step is a class of parasitism cases in which observational success masks off-partition dependence. The canonical example is the latent-driver proxy: a macro-variable aggregating two subsystems driven by a shared upstream process can satisfy observational screening off because the driver is a common cause of the proxy’s present and its future, so that the proxy’s current state carries exactly the upstream information its next state turns on. Under passive observation the proxy looks transition-autonomous. Perturb the proxy directly with an admissible intervention while leaving the drivers intact, and hidden heterogeneity reappears: within-class micro-differences now matter for the ensuing macro-transition. Section 3.3 makes this case explicit as a countermodel rather than an intuition, since the argument turns on the claim that a candidate can satisfy the stated admittance conditions in full and still fail transition autonomy. A second case, hidden-feedback parasitism (§3.3.2), shows the same failure signature from a different source, when fast off-partition dynamics rather than a shared upstream driver sustain the apparent autonomy. The full development in §3.3 shows why these cases decide the criterion.

Throughout this paper, *transition autonomy* names the property the criterion tests: the candidate partition’s own dynamics close under admissible perturbation, so within-class micro-detail adds no predictive power. *Gerrymandered* then means dynamically incoherent: a partition that fails transition autonomy by requiring persistent within-class micro-bookkeeping, not merely one that is intuitively disjunctive or aesthetically odd. *Parasitic* means that a macro-variable’s apparent autonomy is borrowed from upstream structure rather than realized in the candidate partition’s own transition organization.

Three claims of different strength are in play, and readers who grant one but not the next can take the paper’s contribution in corresponding part. The weakest is that closure is a necessary anti-gerrymandering constraint on serious macro-ontology claims. The middle claim is that, given the structural realist and interventionist commitments stated below, closure under admissible intervention is the appropriate criterion for robust macro-level commitment. The strongest is programmatic: rainforest realism is naturally read as a structured space of computationally closed coarse-grainings rather than as a binary inventory of what is, or is not, real. Sections 2 through 4 develop the first two claims; the structured-space reading is offered as a programmatic upshot in Section 6, with the formal richness of that space in any given domain to be cashed out as the closure discipline matures.

The argument is philosophical, not a methods paper in disguise. It provides a criterion and a disciplined protocol for applying it, while estimator selection, finite-sample behavior, and domain-specific implementation remain downstream methodological tasks.

1.1 Novelty and Positioning

The novelty claim has three parts.

1. Recovering screening off’s modal content: the contribution is not a second condition layered on top of screening off but a correction to how screening off is read. Conditional independence admits an observational reading, on which micro-detail

is redundant given the macrostate across the system’s actual trajectory, and a modal reading, on which that redundancy survives admissible intervention. These readings coincide in the central cases and come apart exactly where apparent macro-autonomy is borrowed from off-partition structure. Franklin and Robertson’s criterion is retained in full; the claim is that their stated test, like the screening-off diagnostics it draws on, fixes only the observational reading, whereas the reading that carries ontological weight is the modal one. On the modal reading, screening off and transition autonomy are not two properties but one: to screen off micro-detail from the macro-future under admissible perturbation just is to carry autonomous transition structure at that grain. Closure is the admissibility-indexed test that recovers this modal content and thereby distinguishes autonomous macro-variables from parasitic ones. Their account is valuable precisely because it frames rainforest admission without treating reducibility by itself as disqualifying (Franklin and Robertson 2024). The reading is not a peculiarity of one paper. Meyer generalizes the same package beyond the special sciences and states the point at issue explicitly, defining dependence as unconditional relevance and *autonomy* as conditional irrelevance, $P(E_2 \mid E_1 \ \& \ B) \approx P(E_2 \mid E_1)$ (Meyer 2026, p. 19). Autonomy is there identified with a probabilistic condition on an unindexed distribution, which is precisely the identification this paper argues underdetermines what it is meant to secure.

2. Beyond formal closure results alone: recent work distinguishes informational, causal, and computational closure as formal diagnostics, within the formal settings studied, (Rosas et al. 2024), but does not say when those diagnostics warrant ontological commitment or when commitment should be withdrawn. Those diagnostics are moreover defined over the observed process, and their authors are explicit that the counterfactual reading requires the underlying system to be fully observed (Rosas et al. 2024, Sec. III), which is why they secure the observational reading rather than the modal one on their own. The criterion here uses those results for a more explicit philosophical purpose: a condition on robust macro-level standing with explicit intervention indexing, graded verdicts, and downgrade conditions.
3. Beyond effective realism without downgrade discipline: recent work in ontic structural realism allows regime-dependent, scale-relative ontology, at the level of metaphysical package rather than an individuation criterion, (Ladyman and Lorenzetti 2024), but leaves open what operationally distinguishes robust admission from cases that should be downgraded or withheld. Closure supplies that discipline.

The positioning is deliberately dialogical. With Dennett, the paper keeps compression realism while refusing to let compression settle admittance by itself (Dennett 1991). With Franklin and Robertson, it agrees that screening off and novelty are indispensable but argues that they do not yet separate autonomous from parasitic macro-variables (Franklin and Robertson 2024). With Ladyman and Ross, it keeps the rainforest ambition while insisting on an explicit transition criterion for robust macro-level commitment (Ladyman and Ross 2007). And with Ladyman and Lorenzetti, it takes ontology to be scale-relative and regime-dependent while remaining answerable to objective modal structure (Ladyman and Lorenzetti 2024); the present paper then uses interventions as diagnostics of structural autonomy rather than as metaphysical

primitives. One consequence of intervention indexing is that the closure criterion can diverge from purely observational diagnostics: a macro-variable can achieve predictive sufficiency by tracking an off-partition driver rather than by instantiating autonomous transition structure (§3.3).

Several neighboring traditions are engaged where they become dialectically active: Simon and Ando on near-decomposability and aggregation (Simon 1962; Ando and Simon 1961), Wimsatt on non-aggregativity and reductionist bias (Wimsatt 2000), mechanistic accounts of organized explanation (Machamer et al. 2000; Craver 2007), Rosas et al. on formal closure diagnostics (Rosas et al. 2024), causal-emergence work (Hoel et al. 2013; Dewhurst 2021), and recent efforts to tighten or reframe pattern realism and adjacent structuralist debates (Millhouse 2022; Wallace 2024; Jiang 2024, 2025; Ladyman 2026; Meyer 2026). Each is addressed at the point where its specific contribution bears on the argument. The narrower claim is that even the strongest of these views leaves open an admittance question that closure helps make more explicit and more operational.

1.2 Intervention, Structure, and the Causation Worry

Ladyman and Ross are skeptical that causation belongs in fundamental metaphysics, and recent effective ontic structural realism continues that broadly structuralist posture (Ladyman and Ross 2007; Ladyman and Lorenzetti 2024). By contrast, the criterion defended here is explicitly intervention-indexed. That can look like a retreat from structural realism to a Woodwardian metaphysics of causation. It is not.

Intervention does not function here as an ontological primitive. It functions as a diagnostic probe for structural autonomy. Admissible intervention is the most disciplined test of whether a candidate macro-partition preserves stable relational organization under perturbation. What the criterion tracks is still structure. The intervention test asks whether that structure is autonomous at the candidate grain or whether the apparent autonomy was borrowed from elsewhere. Woodwardian intervention tests typically operate within an accepted variable scheme; the criterion here asks the prior question of when a candidate coarse-graining has earned the right to figure as a macro-variable at all.

The proposal is compatible with effective ontic structural realism because Ladyman and Lorenzetti explicitly allow ontology to be regime-dependent and scale-relative while remaining realist about the structures discovered there (Ladyman and Lorenzetti 2024). The present account agrees, but adds a sharper admittance condition: a scale-relative structure earns robust ontological standing when its transition profile is stable under the admissible perturbations that define the regime. In that sense, intervention is used here to supply operational discipline for effective realism rather than to compete with it. Their claim is not that ontology collapses into convenience, but that ontology is always effective ontology at some scale and under some modal profile (Ladyman and Lorenzetti 2024); closure adds a stricter account of when such scale-relative commitment should be robust, qualified, or withheld.

The paper requires intervention-invariance as evidence that a candidate partition tracks stable organization across physically real, regime-preserving perturbations to

boundary conditions, couplings, or control channels, not interventions as metaphysical glue. That is a methodological route to structural stability, not a retreat to fundamental causal ontology.

There is a principled reason to resist intervention-indexing that deserves a direct answer, since it is not the familiar complaint that interventions are merely difficult to perform. Meyer defends the conditional-probability approach partly on the grounds that it avoids spatiotemporal presuppositions and preserves metaphysical neutrality, which matters where the spatiotemporal and causal structure that interventionist tests presuppose is itself what is in question, as in quantum gravity (Meyer 2026). That argument is a good one, and the reply is a concession rather than a rebuttal. Where no admissible intervention class can be specified, the criterion does not deliver a robust verdict; it delivers an indeterminate one, which is the downgrade discipline working as intended rather than failing. The scope claimed here is induced closure in systems that have a control ecology (§1.3), and in that scope the observational reading is not neutral between autonomous and parasitic candidates, which is what the argument below turns on. Meyer’s desideratum and this criterion are therefore answering different questions: his asks what travels into domains where intervention talk may not apply, and this one asks what settles admission where it does. The cost of the neutral reading is that it cannot discriminate exactly where discrimination is possible.

1.3 Scope and Non-Claims

Closure comes in two forms, stipulated and induced. Stipulated closure holds when autonomy is fixed by constitutive rules, as in chess or formal arithmetic; once the rules are set, transitions and consequences follow by construction. Induced closure holds when autonomy is discovered through dynamical screening-off in the physical world. The restriction to induced closure is principled rather than ad hoc. Induced closure asks whether transition autonomy is sustained by physical and organizational structure independent of whether anyone stipulates that it should be; stipulated closure asks only whether a consequence-relation holds within a rule-system. Institutional kinds are hybrid, combining constitutive rules with an implementation substrate, and the criterion tests whether the induced component sustains the stipulated one under admissible perturbation. That is why legal-tender cases can fail closure even when the rules persist (§4.5): what degrades is the induced infrastructure, not the stipulated rule. The scope here is accordingly restricted to induced closure in spatiotemporal systems, leaving aside broader debates about abstract objects and a complete metaphysics of levels; where those debates arise, they are handled only to protect the central claim from predictable misreadings. Still, the criterion is stated over state-transition structure: spatiotemporal dynamics are the paradigm case, not the only intelligible format in which closure questions can be posed.

Methodologically, the paper does not depend on the strongest anti-analytic rhetoric sometimes associated with naturalized metaphysics. The claim is narrower: closure is a usable philosophical constraint that can be motivated by dynamical and coarse-graining practice, whether or not one accepts every meta-metaphysical commitment in the broader *Every Thing Must Go* (ETMG) program (Ladyman and Ross 2007).

The contrast class is correspondingly focused. The paper rejects the view that compression and prediction are sufficient for realist commitment, but it also rejects a fallback to utility alone. The claim defended here is closure realism: compression and projectibility identify serious candidates, while robust macro-level commitment requires transition autonomy under fixed regime, horizon, and admissible intervention class. The paper has three argumentative layers: Section 2 states the criterion and its admissibility discipline; Section 3 gives the exact benchmark in Markov settings; Section 4 extends the same criterion to non-ideal settings as a graded protocol. Section 5 addresses the strongest objections. Section 6 summarizes the gains and limits.

2 The Criterion: Closure Under Admissible Intervention

2.1 Criterion in Plain Language

Closure can be stated without heavy formalism. A candidate macro-variable passes when knowing its current macrostate is enough to determine its macro-future over a specified horizon and intervention class. If two microstates inside one macrostate produce different macro-transitions, closure fails at that grain.

Static patterns are not exceptions; they represent dynamical regimes where micro-fluctuations are screened off rather than absent, so persistence is itself a type of macro-transition that falls squarely within the criterion. What matters is transition sufficiency, not descriptive fit. A representation can summarize trajectories elegantly and still fail to support robust macro-level commitment if it requires persistent within-class micro-bookkeeping to preserve forecast quality.

Since the formal machinery evaluates partitions of state spaces rather than physical things directly, an explicit bridge is needed. On the structural realist commitments defended below (§2.12), realist commitment at the macro level tracks stable relational and causal organization at the relevant grain; the partition identifies that organization.

Four levels need to be kept apart to avoid recurring misunderstandings.

1. World dynamics: the implemented process itself.
2. Pattern type: stable transition structure supported by that process.
3. Pattern token: a concrete instance under specific boundary conditions.
4. Representation: a model, equation, classifier, or narrative that tracks the pattern.

A representation can fail while the pattern type remains real. A token can fail while the type remains robust. A token can also succeed while the type is weak, for example in narrow calibration windows. Closure claims here are primarily type-level claims under specified regimes, then secondarily claims about token reliability under perturbation.

Three quick clarifications forestall predictable misreadings. Closure is not a test for every sense in which a pattern might be real; it is a test for robust macro-level standing under declared constraints, so weaker senses of pattern reality remain available to candidates that do not meet it. Closure is not a demand for perfect transition autonomy; the graded verdict structure handles partial cases through qualified and

indeterminate categories. And closure is not a replacement for the special-science kinds it evaluates; most will earn qualified rather than robust standing, and that is the intended output rather than a failure of the criterion.

2.2 Which Reading of Screening Off Closure Secures

A natural rejoinder is that screening off, properly understood, already excludes parasitic candidates, so closure merely renames an existing criterion. The concession is worth making directly. Closure is not a rival to screening off, and the argument does not require that Franklin and Robertson overlooked causation or meant something weaker than they said. What closure supplies is not a further condition but a determinate reading of an existing one, together with the test that secures it. That is a substantive contribution only because the two readings come apart, and come apart on candidates that matter.

1. Weak reading: observational conditional irrelevance. A variable's value renders further micro-detail probabilistically redundant for the target given the system's actual dynamics. This is a statement about the observational joint distribution, and it is what statistical screening-off diagnostics test.
2. Strong reading: stability under admissible intervention. The same conditional independence continues to hold under the interventional distribution generated by admissible perturbations that target the candidate partition directly, so that screening off is tracking autonomous structure rather than a shared source.

The claim here is that established screening-off diagnostics generally secure only the weak reading, while the strong reading is what warrants robust macro-level commitment. Closure operationalizes the strong reading by adding the admissible intervention class as a parameter of the test. The two readings agree in the central cases and diverge when upstream structure lies outside the modeled state space, or when admissible interventions target the candidate partition directly while leaving that structure intact. The latent-driver proxy developed in §3.3 is the canonical wedge; the hidden-feedback case (§3.3.2) supplies a second wedge whose borrowed stability has a different source, so the divergence is not tied to one peculiar configuration.

The reply to the rejoinder is therefore to accept it and press it further. Screening off properly understood is modal screening off, and modal screening off is transition autonomy: there are not two properties here but one, described once in the vocabulary of conditional independence and once in the vocabulary of dynamics. What does not follow is that the criterion is idle. A reading that no stated test discriminates cannot adjudicate admission, and the parasitism cases show that the undiscriminated readings return opposite verdicts on candidates the debate cares about. The contribution is the test and the discipline that goes with it, not the discovery of a new property. The intervention-indexed reading is offered as one principled way of making the intended modal content operationally discriminating, which is why the argument is best understood as an explication of rainforest admission from within rather than an objection to it.

2.3 Formal Statement

Let micro-process be X_t and candidate macro-process be $Z_t = g(X_t)$. For horizon L , closure asks whether Z_t is sufficient for forecasting Z_{t+1}^L under a fixed regime and admissible intervention class.

The horizon object here is the L -step path distribution, not only one-step prediction. One-step tests can be used as diagnostics, but realist claims are indexed to the declared horizon target.

Informationally, the predictive side asks whether adding X_t beyond Z_t yields material gain for macro-future prediction. Interventionally, the causal side asks whether macro-transition laws remain stable across admissible perturbations without needing within-class micro-identification.

These are not competing tests. They are two readings of the same target: transition autonomy at the macro level.

2.4 Why This is More Than Compression

Compression is necessary but not sufficient for robust macro-level commitment, because a compressed code can hide transition-relevant heterogeneity and still score well on narrow tasks. Closure blocks that loophole: does hidden heterogeneity matter for onward macro-behavior?

Closure therefore functions as an anti-gerrymandering condition, rewarding coarse-grainings that carry stable transition structure under declared constraints rather than convenient coding alone.

Put differently, the account accepts Dennett's claim that compression success is evidence of objective patterning while denying that compression success is the full ontological test (Dennett 1991), and it accepts Ladyman and Ross's concern that real patterns should be projectible and structurally disciplined while adding an explicit transition criterion for adjudicating borderline cases (Ladyman and Ross 2007). The aim is not to replace those insights but to make their realist force less permissive.

This also answers the Haugeland-style pressure point: the transition-autonomy condition determines when pattern talk has ontological force rather than merely descriptive convenience (Haugeland 1998).

2.5 Closure and the Primacy of Physics Constraint

Ladyman and Ross's Primacy of Physics Constraint is often read negatively, as a warning against ontologies that float free of physics. But it contains a second, equally important pressure: macro-candidates must not only be physically compatible but must also avoid redundancy with lower-level description in the realism-relevant sense (Ladyman and Ross 2007). The missing question is what that non-redundancy amounts to once one gives up the idea that only the micro-level is fully real.

Closure gives that vague requirement operational content. A macro-candidate is non-redundant when micro-differences within one macroclass do not alter macro-transitions under the declared regime, horizon, and admissible intervention class. That is a stronger claim than descriptive usefulness, and a different claim from mere compression. A macro-description can be shorter, more tractable, and even explanatorily

attractive while still remaining redundant in the sense that its apparent autonomy depends on continued micro-level repair.

The latent-driver pressure case brings the point home. A proxy can look perfectly projectible under passive observation because upstream structure keeps its trajectories aligned, yet turn out to be transition-parasitic: perturb the candidate directly with an admissible intervention, and within-class micro-differences matter again. The apparent macro-autonomy disappears. Closure therefore sharpens the Primacy of Physics Constraint’s non-redundancy demand by making it a question about transition structure rather than about descriptive convenience.

Read this way, closure is not an external supplement imposed on rainforest realism but a way of saying more exactly what it takes for a higher-level pattern to be physically respectable without collapsing back into lower-level bookkeeping. The realism-relevant discriminator is causal autonomy under admissible perturbation, not microphysical smallness by itself.

2.6 Admissibility and Hierarchical Evaluation

Closure is always relative to an intervention class, and the standard objection is that this invites circularity: who picks the class? The answer is procedural rather than metaphysically ambitious. Any serious macro-ontology has to answer the same admissibility question, since any view that evaluates macro-variables has to say which perturbations, regimes, and horizons are relevant. Closure does not invent that burden; it makes the corresponding choices public, comparable, and downgrade-sensitive. Domain interests help determine which control channels are worth testing, but those choices are constrained by the system’s dynamics, the target regime, and the physically available modes of perturbation. What matters for objectivity is that admissibility classes are declared in advance, cross-compared, and audited. Fragility across nearby defensible admissibility choices is itself evidence for downgrade. On this view, robust macro-level standing turns on stability of macro-transition structure across the perturbations the regime can physically sustain, not merely on fit to observed trajectories.

This procedural discipline is the framework’s main defense against gerrymandering. Even when admissibility classes are guided by domain interests, a candidate does not earn robust standing simply because one carefully chosen intervention class makes it look good. If the verdict collapses under nearby defensible admissibility specifications, that fragility is itself objective evidence against a strong realist commitment. The point is not to pretend that human interests disappear. It is to ensure that they cannot silently protect a favored partition from downgrade.

Admissibility is fixed upstream of commitment verdicts by three constraints:

1. Epistemic admissibility: interventions are measurable and implementable by bounded agents.
2. Dynamical admissibility: interventions preserve the target regime class.
3. Explanatory admissibility: interventions target variables with cross-context generalizability and non-trivial counterfactual reach, rather than one-off manipulations tuned to a single episode.

These constraints are physically anchored in available control channels and implementation structure, not in analyst preferences about favored ontologies. The explanatory constraint is set before closure verdicts and does not assume in advance which candidate partition will prove stable. This makes the interventionist notion of a “possible experiment” concrete and bounded rather than leaving it at conceptual possibility (Woodward 2015), and aligns with Woodward (2021)’s proportionality demand on variable choice. Explanatory admissibility is stronger when its control channels already exist in the domain and replicate across sites, weaker when success depends on narrow one-off tuning. Domain expertise shapes regime identification and control-channel selection, but that shaping is declared, auditable, and separated from partition scoring.

Pearl’s $do(x)$ surgery is defined mathematically regardless of physical feasibility, which is productive for causal inference (Pearl 2009). Robust macro-level commitment requires tighter constraints: a partition that closes only under interventions no bounded agent could implement without destroying the target regime does not support the intervention-guiding structure that warrants realist commitment. The restriction concerns which of Pearl’s outputs carry ontological weight, not the framework itself.

This connects back to the foundational claim above. The admissible class is not arbitrary: a system’s boundaries are probed constantly by naturally occurring perturbations (thermal noise, diffusion gradients, mechanical impacts), and a cell membrane screens molecular fluctuations whether or not an experimenter is watching. These perturbations are exogenous to the candidate partition even when internal to the larger system, so the test inherits the exogeneity that standard interventionist accounts demand without requiring human agency to secure it.

Evaluation order is fixed by the canonical selection discipline (§4.3): intervention classes are fixed from independent physical constraints and established control practice before closure is scored, and post hoc revisions restart the evaluation. If verdicts remain highly sensitive across nearby defensible admissibility specifications, commitment should be downgraded.

Taken together, the discipline can be stated as a short acceptability test. An admissibility class is acceptable only when:

1. interventions use physically available control channels;
2. interventions preserve the regime being evaluated;
3. no intervention is one-off or directly partition-protective;
4. verdicts are checked across nearby defensible alternatives.

Classes that fail any of these conditions cannot carry the evidential weight of a robust verdict.

2.7 Regime Individuation and Circularity

If regime individuation already commits to which macro-variables matter, the protocol risks circularity: closure testing presupposes regime identification, but regime identification might presuppose partition outcomes. The reply has a structural and a procedural part. Structurally, the criterion is designed to sort cases into three classes rather than to force robust verdicts everywhere. In the first class, regime specification

is uncontroversial across candidate partitions, and the criterion adjudicates robustly. In the second, regime specification is contested but iteration stabilizes under the selection discipline (§4.3), and reflective equilibrium converges on a verdict. In the third, regime specification is contested and iteration does not stabilize, so the criterion issues an indeterminate verdict; this is the discipline working as intended, not the criterion failing. Procedurally, defensible regime specifications draw on quantities characterizable without presupposing which candidate partition will succeed. Temporal parameters, energy scales, flow rates, and enforcement intensities are typically identifiable through physical indicators that competing candidate partitions can agree on before either is evaluated. A running example will help here and at two later points, and it can be stated in a line. Consider two rival partitions of weekday commuter traffic in a metropolitan corridor: P_{lane} , which tracks density and flow by lane segment, and P_{tod} , which tracks flow volume in clock-time buckets calibrated to typical commuter patterns. Both are choices a real modeler might defend, and their in-sample one-step fit can be similar. The regime is characterized by clock schedules, total vehicle counts, and road network topology, all quantities that investigators favoring either partition can agree on before either closure test is run. Regime specifications of this kind are anchored in lower-level or cross-cutting physical indicators rather than in the outcome of the very test they enable. Regime specifications should be grain-neutral across competing candidate partitions: a regime can itself be a macro-level description while still not presupposing which finer-grained partition will close within it.

Independence is imperfect in some domains, particularly biology and social science, where macro-variable choice and regime characterization can be entangled (psychiatric diagnostic categories, species concepts: the variables that pick out the regime are themselves candidates whose macro-autonomy is under test). This is where the criterion earns its keep rather than stumbles: when alternative, independently motivated regime specifications produce divergent closure verdicts, the right result *is* downgraded to at most qualified, because the evidential situation does not yet support a stronger commitment. The criterion is doing its intended work when it refuses robust standing to candidates whose apparent autonomy is parasitic on the choice of regime characterization.

Take a contested case as a stress test, sketched schematically rather than settled here. Suppose major depressive disorder is approached under three independently motivated regime characterizations: a symptom-cluster regime tied to diagnostic instruments, where intervention evidence on pharmacological and psychotherapeutic levers might support partial closure within a bounded clinical horizon; a neurobiological regime indexed to specific circuit dysfunctions, where the same label may fragment across heterogeneous mechanisms; and a social-functional regime indexed to role-impairment trajectories, tracking a third transition profile. If the case has that shape, the criterion returns qualified standing, with explicit downgrade for any commitment that depends on which characterization is accepted upstream. Whether it has that shape is a question for philosophy of psychiatry, and nothing here settles it. The point is what the criterion does with a kind whose regime is contested, not a verdict on this kind. Refusing to launder a contested kind into robust standing is the discipline working as intended, not evasion.

Regime individuation and partition evaluation are iterative rather than viciously circular: initial regime characterizations are provisional, and closure testing can refine them. The process stabilizes into reflective equilibrium in Goodman’s sense (Goodman 1983): if iteration does not stabilize across nearby defensible starting specifications, the result should be treated as indeterminate rather than forced into a robust verdict. Here stabilization means that comparative ordering across independently specified candidate partitions stops shifting under small, defensible protocol variations; it does not presuppose one privileged grain in advance. Iteration is governed by the same selection discipline (§4.3) as initial scoring: each revision restarts the evaluation rather than silently inheriting prior results. The framework sometimes relocates disagreement rather than resolving it, and that is the intended outcome in hard cases. The aim is not to guarantee a unique answer in every domain but to distinguish cases where robust commitment is warranted from cases where the evidential situation supports only qualified or indeterminate standing.

“Regime dissolution,” the event that marks the outer edge of a regime, is not partition-relative. It is characterizable by cross-cutting physical indicators that do not presuppose the winning partition: energy-scale crossings, phase boundaries, control-response failure at established channels, and breakdowns in the separation of timescales that defined the regime. These indicators register whether the background dynamics have changed, not whether a favored partition continues to score well.

2.8 Pattern Reality Versus Robust Macro-Commitment

A pattern can be real in a weaker sense without satisfying the stronger criterion defended here. A representation can capture regularities that are descriptively useful while still failing closure under admissible interventions.

The closure criterion is stricter. It targets robust macro-level standing, not any and every projectible summary, so the view can preserve a generous attitude toward pattern detection while denying robust ontological standing to high-leak candidates. Indeterminate standing is a refusal to overstate commitment when evidence does not stabilize, and a candidate can be worth tracking under a verdict short of robust.

That distinction also clarifies the paper’s relation to Franklin and Robertson. Their screening-off condition is well suited to identifying when a macro-variable has become explanatorily and nomically serious (Franklin and Robertson 2024). The claim here is narrower and stricter: some variables that are serious enough to survive that filter still fail transition autonomy and so do not yet qualify for robust realist commitment.

So the dialectical burden is limited: the paper need only show that non-closed representations do not meet the stronger standard defended here, not that they are worthless.

2.9 Admissibility Disputes

Admissibility disputes are expected, especially across domains. The adjudication rule is to compare candidate intervention classes by what they physically permit and whether they preserve the same target regime. Boundary-pressure perturbations can be admissible for storm dynamics, while molecule-by-molecule remote rewriting is

not. In institutional settings, changing enforcement intensity can be admissible, while instant arbitrary rewriting of all agent commitments is not. The ontological consequence is direct. A storm satisfies closure under boundary perturbations because its pressure organization and rotational structure determine macro-transitions without tracking individual molecules. Under molecule-by-molecule rewriting, the same candidate fails, because admissibility now permits interventions that reach inside the macroclass and exploit within-class differences. The candidate has not changed; the admissible control channel has, and the verdict changes with it. Admissibility must therefore be declared and justified before scoring, not adjusted afterward to protect a preferred outcome.

The issue is distinct from regime individuation. Regime individuation asks which background dynamics define the target setting; admissibility individuation asks which intervention classes count as ontologically relevant probes within that setting. Tying explanatory admissibility to available control channels and implementation structure, rather than to interventions that happen to vindicate a favored partition, answers the circularity worry here as well.

Front-end specification of regime, target, and admissible control channels is often interest-guided in a perfectly ordinary way. That does not make closure verdicts interest-relative once those specifications are fixed and openly defended. Where they fail to stabilize across defensible alternatives, the right result is downgrade or indeterminacy rather than forced realism. A short template grid (boundary nudges, coarse actuator controls, policy levers) can make admissibility comparison public and repeatable in domains that lack an established control lexicon.

One guardrail is non-negotiable. Interventions that directly target preservation of partition labels, rather than system variables, are inadmissible because they trivialize closure by design. The test concerns whether transition structure is autonomous under physically meaningful controls, not whether labels can be protected by stipulation.

When two intervention classes are both admissible and target the same regime, the criterion permits plural testing. Verdicts should report which class was used and how sensitive results are across nearby defensible classes; sensitivity to a narrowly tuned class is grounds for downgrade. In the commuter corridor of §2.7, ramp metering and speed-limit controls count as admissible because they are physically available control channels in the regime, while vehicle-by-vehicle remote rewriting does not, not because that would embarrass P_{lane} but because it is not part of the regime’s actual control ecology.

2.10 Canonical Criterion Statement

The core criterion:

A candidate coarse-graining $Z = g(X)$ qualifies for robust macro-level standing in regime when, under fixed horizon L and admissible intervention class $\mathcal{I}$, macro-transition structure is autonomous up to declared tolerance, so within-class micro-differences do not materially improve macro-future prediction or intervention-guided control.

Candidate partitions must be specified independently of the particular sample trajectory used to score closure, for instance by a pre-declared construction rule or

learning objective fixed before evaluation. Otherwise closure collapses into bespoke encoding, where any dataset can be made to look autonomous by post hoc recoding.

In exact Markov settings, strong lumpability is a sufficient benchmark realization of this criterion. In non-ideal settings, leakiness-centered diagnostics estimate distance from that ideal under fixed constraints.

2.11 Predictive and Interventional Closure

Predictive and interventional closure should be distinguished but not separated. Predictive closure concerns whether macrostate information screens off transition-relevant micro-detail for macro-future forecasting; interventional closure concerns whether macro-transition structure remains stable when the system is probed.

The evidential relation between them is asymmetric. Strong interventional closure typically implies predictive adequacy for the same target and horizon, while predictive adequacy alone can persist in cases where interventional stability fails under distribution shift. For that reason, the criterion treats predictive evidence as important but not final.

This asymmetry also clarifies burden of proof. Claims to robust macro-level standing need either direct interventional support or compelling indirect evidence that tracks intervention-relevant invariance. Otherwise, the responsible verdict is qualified or indeterminate.

Indirect evidence can still be substantial: natural experiments, intervention-like policy shifts, quasi-experimental contrasts, and systematic failure under exogenous disruption can all reveal whether a candidate partition preserves its own transition structure across admissible changes. These sources do not collapse the distinction between predictive and interventional closure. They are useful precisely because they bear on intervention-relevant invariance rather than on in-sample fit alone.

The point matters especially in biology, economics, and the social sciences. In those domains, direct manipulation is often sparse, ethically constrained, or expensive. The ontological target nevertheless remains interventional closure. Where the evidence for that target is thin, robust verdicts should be rare. That is a feature of the graded verdict structure rather than a retreat to observational permissiveness.

2.12 Why Closure Carries Ontological Weight

A reviewer can still ask why transition autonomy should count as robust macro-level standing rather than as a mere success condition for modeling. The answer depends on the paper’s explicit commitments. Under structural realism, what ontology should track is stable relational and causal organization rather than intrinsic micro-identity as such. Under interventionism, what counts as causally relevant structure is structure that supports stable manipulation and control.

This fits the regime-dependent, scale-relative orientation defended by effective ontic structural realism. The proposal does not ask for a return to one privileged level or to thick intrinsic objects, but for a sharper criterion of when a scale-relative structure has earned commitment, and when it should be downgraded, within a broadly structural realist framework (Ladyman and Lorenzetti 2024).

The meta-philosophical stance here follows what Woodward calls the methodological path to ontology: causal-interventionist tools are used to determine what earns realist commitment, rather than seeking independent metaphysical “truth-makers” or “grounds” that stand apart from the practices of manipulation and prediction (Woodward 2015). On that path, scientific ontology disciplined by closure is the only ontology required for commitment verdicts at this grain. This is a consequence of the stated commitments rather than an independent argument against those who reject them. A grounding theorist who holds that stable dynamical autonomy under admissible intervention still stands in need of a further metaphysical foundation is not making a mistake by the lights of their own view; they are declining the methodological path at the outset, and the disagreement is therefore about that path rather than about closure. Readers who take it will find that closure settles what commitment requires; readers who do not can still take the anti-gerrymandering result of §3.3, which does not depend on the ontological reading.

A persistent worry is that prediction and interventional control are merely epistemic lenses, leaving the underlying ontology untouched. But this confuses the test with what is being tested. Prediction and intervention are diagnostic tools that reveal pre-existing statistical structure. The conditional independencies that allow a macro-state to screen off micro-details exist whether or not anyone computes them; they are the terrain, not the lens. Locality, finite signal propagation, and thermodynamic limits create these independencies as physical facts. Our predictive models succeed when they align with this objective causal topology, and fail when they do not. Closure is therefore an ontological criterion because it identifies where the world’s phase space objectively decouples, not just where our minds prefer to chunk it. Under structural realism, such decouplings are ontological because they are the stable relational structure realist commitment was supposed to track in the first place, not because they license any particular representation. A criterion that identifies them is therefore picking out what the ontology consists in at that grain, not describing a consequence of it.

Given those commitments, closure functions not as a convenient heuristic layered on top of ontology but as the criterion that identifies when a candidate macro-description tracks stable structure at the level where explanation and intervention are being assessed. That is what lets the view remain compatible with microphysical completeness while still defending non-redundant macro commitment. The paper is not smuggling ontology out of modeling success; it is defending a specific thesis about which kind of modeling success is ontology-relevant, namely transition autonomy under admissible perturbation.

If one asks what the rainforest amounts to once admission is disciplined this way, the most natural reading on this view is not a loose plurality of levels but an ordered space of computationally closed coarse-grainings. Rosas et al. provide the formal architecture for talking about such relations among coarse-grainings or predictive machines (Rosas et al. 2024). The reading is offered programmatically: an ordered space of this kind gives the best interpretation of disciplined rainforest realism under the paper’s stated commitments, while the strength of its formal properties (full lattice structure, completeness across regimes) remains a domain-conditional question to be settled as

the closure discipline matures. Admission, qualification, and downgrade are therefore not floating labels imposed from outside the formalism; they locate a candidate within a structured space of more and less stable closure relations.

2.13 Structural Contrasts

Pearl’s notion of modularity is the nearest structural contrast. Modularity is a property of a causal model whose variables have already been accepted; closure asks the prior question of whether a candidate coarse-graining has earned the right to appear as such a variable at all (Pearl 2009). Mechanistic accounts make the same point in a different idiom: organized entities and activities can support stable explanation, but that by itself does not show that a proposed macro-variable has been carved at the right place (Machamer et al. 2000; Craver 2007). Closure supplies that prior sorting question.

The proposal is also not vulnerable to Newman-style triviality pressure in structural realism. Closure requires modal and dynamical transition stability rather than extensional fit under redescription, so arbitrary recodings typically fail once that stability is demanded under declared constraints (Ainsworth 2009, pp. 141–145). A closure-supporting partition is therefore more than a merely “good variable” in some deflationary sense: it tracks objective modal structure under admissible perturbation, whereas a merely convenient representation need not.

A related body of work on asymptotic explanation and universality classes bears directly on closure. Batterman argues that certain explanatory structures emerge at the interface between scales and resist reduction to any single descriptive level (Batterman 2002). That observation is compatible with this account: universality classes show why heterogeneous micro-configurations can yield stable macro-behavior despite wide micro-variation. Batterman’s focus is explanatory structure; closure adds a further commitment test that determines when such inter-scale structure warrants realist commitment rather than merely explanatory reliance.

The criterion therefore does not equate model performance with ontology. It asks whether the represented transition structure is autonomous, not whether one representation currently performs well.

2.14 Closure, Underdetermination, and Commitment Discipline

One remaining concern is underdetermination. Even after the criterion is stated, a reviewer may argue that many distinct coarse-grainings can be tuned to look acceptable on available data. If so, closure might seem to collapse back into model choice rather than commitment discipline.

Three questions need to be kept apart.

1. Which candidate partitions are descriptively serviceable in a dataset?
2. Which candidates remain transition-autonomous under admissible perturbation?
3. Which of those candidates remain stable under modest horizon and admissibility variation?

Underdetermination is strongest at the first question and weaker at the second. It is often weakest at the third. Many candidates can fit observed trajectories. Far fewer preserve counterfactual structure when the regime is probed. Fewer still remain stable across nearby defensible setups.

The account does not claim that every domain yields one uniquely privileged partition. It claims that realist commitments should be constrained by transition autonomy and robustness, rather than by fit alone. When underdetermination persists after those filters, the responsible result is qualified or plural commitment under transparent constraints, not forced monism and not unconstrained relativism.

Once admissibility, horizon, and intervention class are fixed, the question is not simply whether one favored partition closes. The question is which coarse-grainings close, which close only conditionally, and which fail. That answer is what the rainforest becomes under a closure discipline: the structured space of computationally closed coarse-grainings, ordered by refinement and by comparative stability. (Rosas et al. describe the formal result as a “lattice of nested computational structures” (Rosas et al. 2024, p. 2); whether the space has full lattice properties in every domain is an open formal question, but the partial ordering and nesting relations it supplies are what matter philosophically.) Mere membership in that structure is not by itself sufficient for robust macro-level standing. The structured space catalogues the partitions that satisfy closure under the declared constraints; what earns a robust verdict is the partition’s standing once admissibility variation and intervention-robustness checks are applied. Lattice membership is necessary; stability across admissibility and perturbation is what determines whether that membership carries ontological weight.

Three metaphysical clarifications sharpen this payoff. First, the structured space is indexed to a regime. Each regime, with its declared horizon and admissible intervention class, picks out its own structured space of closed coarse-grainings. Cross-regime comparisons are then further relations among those spaces rather than a single global inventory: where regimes themselves nest or overlap, their closure structures inherit corresponding nesting or overlap relations. This preserves the regime-indexing that earlier sections impose without collapsing into a multiplicity of unconnected inventories.

Second, within a single regime, closed partitions scored under independently motivated admissibility classes form parallel sub-structures rather than a single ordering. A candidate that closes under one admissibility class may close, close only conditionally, or fail under another. Cross-admissibility stability (§4.6) is what turns parallel local placements into a robust joint placement; sensitivity across admissibility classes is itself evidence that the candidate’s standing in the structured space is not yet stable.

Third, the ordering relation is a feature of the dynamics under the declared constraints, not a representational choice. Refinement is a relation among partitions of the same microstate space, and comparative leakiness under fixed regime, horizon, and intervention class is a quantity about the system’s transition structure. The analyst declares the constraints; the ordering that emerges given those constraints is read off. Whether that ordering has full lattice structure in every domain remains an open formal question, as noted above, but the weaker partial-order and nesting relations are domain-general enough to carry the philosophical work.

When exact closure is unavailable, leakiness-centered diagnostics estimate comparative distance from closure, and commitment strength tracks the stability of that evidence.

3 Exact Benchmark: Strong Lumpability

The paper’s relation to Rosas et al. can be stated exactly. What is taken from their treatment is the formal diagnostics of informational, causal, and computational closure, and the strong-lumpability benchmark (Rosas et al. 2024). What is added here is an intervention-indexed criterion for when such diagnostics warrant robust macro-level commitment, together with downgrade rules for non-ideal cases. The formal results are read as the ideal limit of the criterion rather than as the criterion itself.

Strong lumpability is introduced as a benchmark, not as destiny: a clean exact case in Markov settings where closure can be stated and checked without ambiguity.

Let partition cells under g be macroclasses. Strong lumpability holds when microstates within the same macroclass induce identical transition probabilities to all macroclasses. This is the standard strong-lumpability condition; cf. the presentation attributed here to Kemeny and Snell (Kemeny and Snell 1960). When this condition holds, macro-transitions are autonomous by construction.

More precisely: for any two microstates x, x' in macroclass C_i and any macroclass C_j , strong lumpability requires $\sum_{y \in C_j} P(y \mid x) = \sum_{y \in C_j} P(y \mid x')$. The induced macro-kernel is then well-defined independently of representative choice, so the macro-process is itself Markov. A partition that fails this condition mixes transition-heterogeneous microstates and therefore lacks macro autonomy.

The Markov condition does not assume history is irrelevant; it assumes only that the relevant state description has been fixed. In memory-bearing systems, that description may need to be history-enriched before the closure question is asked. That does not trivialize non-Markovian cases or the formal difficulty of establishing lumpability in them; it states the standard point that the closure test applies to the least state description that properly bounds the relevant memory.

Computational mechanics puts the same point in sharper language. For any macro-process Z , two prediction machines can be defined. The ε -machine is the minimal model that predicts Z ’s future from Z ’s own past, using only macro-level information (Shalizi and Crutchfield 2001). The v -machine is the minimal model that predicts Z ’s future from the full micro-past X , using everything available (Rosas et al. 2024). Closure holds when these two machines are equivalent: the ε -machine already captures everything the v -machine knows about macro-futures, and extra micro-information is redundant for the macro target. This gives an exact ideal where closure is not approximate. The same formal point is developed in Rosas et al.’s treatment, where strong lumpability corresponds to the exact case in which the macro-level causal-state machine coincides with the micro-conditioned prediction machine, so that no residual micro-information adds predictive leverage (Rosas et al. 2024, Thm. 3 and Prop. 4, pp. 27–28). The present account reads their formal result as the ideal limit of the criterion rather than as the criterion itself.

The limitation is equally clear: exact strong lumpability is uncommon in open, noisy, and path-dependent systems. That rarity does not undermine the criterion itself.

3.1 Minimal Formal Illustration

The anti-gerrymandering role can be shown with a minimal symbolic example. Let microstates be $\{x_1, x_2, x_3, x_4\}$ with transition rows:

$$\begin{aligned} P(x_1, \cdot) &= (\alpha, \beta, 0, 0), \\ P(x_2, \cdot) &= (\alpha, \beta, 0, 0), \\ P(x_3, \cdot) &= (0, 0, \gamma, \delta), \\ P(x_4, \cdot) &= (0, 0, \gamma, \delta), \end{aligned}$$

with $\alpha + \beta = 1$ and $\gamma + \delta = 1$.

Partition A groups $\{x_1, x_2\}$ and $\{x_3, x_4\}$. Partition B groups $\{x_1, x_3\}$ and $\{x_2, x_4\}$. In A , members of each macroclass have matching onward profiles to macroclasses, so macro-transitions are autonomous. In B , members of one macroclass differ in onward profiles whenever $\alpha \neq \gamma$, so the macro-label hides a transition-relevant difference.

Partition B can still be made predictive by adding repeated within-class bookkeeping. A partition is *high-maintenance* when reliable macro-prediction repeatedly forces refinement of within-class distinctions, so that yesterday’s macrostate labels must be supplemented by new micro-bookkeeping to retain accuracy. That is exactly the case the criterion excludes from robust macro-level standing. On a fixed sample trajectory, this bookkeeping can make Partition B look nearly as predictive as Partition A . The difference appears under admissible perturbation: A preserves macro-transition structure without renewed microstate tracking, whereas B does not.

Both partitions are definable. Only one preserves transition autonomy. Closure therefore discriminates between legitimate macro-candidates and merely codable aggregates.

3.2 If an Unusual Partition Closes

A common reaction is that a strange disjunctive partition might satisfy the formal condition in a symmetric system. That is correct. On this account, if such a partition supports autonomous macro-transitions under fixed constraints, it counts as real at that grain.

That result is not arbitrariness in disguise: the criterion tracks transition coherence rather than intuitive naturalness. A separate naturalness filter, if wanted, is a further criterion and should be declared as such.

Read alongside Andersen’s argument that structure may be dense rather than sparse (Andersen 2025), multiple partitions closing at different grains is what one should expect if the world’s structural organization is richer than any single privileged carving. What matters is whether a candidate earns admission by closure under stated constraints, not whether it is the only closed partition.

The exclusion claim should therefore be read carefully. The criterion excludes dynamically incoherent, high-leak aggregates. It does not exclude every unusual grouping.

3.3 How the Formal Machinery is Used Here

The mathematical property and the philosophical criterion are different things, and the difference is philosophical work the formal framework does not attempt and does not need to attempt for its own purposes. The central such addition is intervention indexing. Causal-state constructions classify histories by equivalence of future distributions under the observed process, which is primarily an observational-predictive equivalence relation (Shalizi and Crutchfield 2001). The criterion here adds an admissible intervention class as a parameter of the closure test. Realist claims are indexed to what the system does under perturbation, not only under passive observation, which separates the criterion from a purely statistical diagnostic. Rosas et al.’s framework distinguishes informational, causal, and computational closure as complementary formal diagnostics (Rosas et al. 2024), and their Theorem 1 establishes that for spatial coarse-grainings, under the assumptions specified there, informational and causal closure coincide (Rosas et al. 2024, Thm. 1). In that framework *causal closure* is defined as equivalence between the coarse-grainings induced by a process’s ϵ -machine and its v -machine (Rosas et al. 2024, Def. 3), and *information closure* as the vanishing of $I(\vec{X}_t; \vec{Z}_{t+1}^L \mid \vec{Z}_t)$ for all L (Rosas et al. 2024, Def. 4). Both are defined over the observed process. Rosas et al. are themselves explicit about what this does and does not license: causal closure as defined there provides counterfactual guarantees only if the underlying system is fully observed, and otherwise “ought to be understood in the Granger sense, i.e. as a best guess given the available knowledge” (Rosas et al. 2024, Sec. III). The equivalence the theorem establishes is therefore one between two readings of observational structure, not between observational structure and interventional invariance under perturbation external to the modeled process. The criterion here is downstream of that result, and it takes their caveat as its point of departure. It treats those observational diagnostics as strong evidence for closure, but ties realist commitment to a further admissible-intervention stability condition and to the absence of hidden micro-identity bookkeeping. The two diagnostics agree in the central cases and can diverge when relevant upstream structure lies outside the modeled state space, or when admissible interventions target the candidate partition directly while leaving such structure intact. That divergence is not a counterexample to their result. It is the region their own full-observation proviso exempts: where the structure sustaining apparent macro-autonomy lies off-partition, the Granger reading is the operative one, and the counterfactual guarantee lapses at exactly the point where parasitism becomes possible. The latent-driver case developed below is the first wedge; the hidden-feedback case (§3.3.2) shows the same failure signature from a structurally different source.

The graded verdicts the criterion issues should be read in the same spirit. They are not merely epistemic confidence labels. They report where a candidate sits in the space of more and less stable closure relations, which is why a downgrade is a claim about the candidate rather than about the state of the evidence alone.

3.3.1 A Latent-Driver Proxy

The first case is the latent-driver proxy, where observational screening off cannot discriminate autonomous macro-structure from borrowed macro-structure. Consider

a composite variable over subsystems that are pinned by an upstream driver process the partition does not track. What matters is not that a single cause synchronizes the subsystems, which is only one way the structure can be realized, but that the driver process is upstream of both the composite’s present state and its future state. The apparent dependence of the macro-variable on its own past is then underwritten by off-partition structure. Under passive observation, the composite’s macro-state predicts its own future as well as the full micro-state does, because the driver pins the subsystems and thereby makes the proxy’s present state carry exactly the upstream information its future turns on. An ϵ -machine built on the composite finds no residual micro-information worth adding, and screening-off diagnostics register a clean pass. Can the screening-off-plus-novelty package handle this case?

Franklin and Robertson’s criterion asks whether the macro-variable enters into a genuine macrodependency that renders some lower-level detail conditionally irrelevant while contributing something novel at the macro level (Franklin and Robertson 2024, §3). Their screening-off condition has two parts. The dependence part, *unconditional relevance*, requires $P(A \mid LLD) > P(A)$: the lower-level description is relevant to the macro-fact. The independence part, *conditional irrelevance*, requires in its strict form that $P(A \mid B \& LLD) = P(A \mid B)$, so that all the influence of the micro-detail is mediated through the macrostate (Franklin and Robertson 2024, §3.1). Their novelty condition is more demanding than it may first appear, and a weaker reading would make the case below too easy. Novelty requires *distinct* macrodependencies: a binary star system screens off the individual stellar trajectories, yet they deny it admission, because the dependencies describing it “are just the same nomic dependencies that describe the trajectories of the two individual stars” (Franklin and Robertson 2024, §3.2). Novelty is therefore not satisfied merely by a dependence being statable at the macro grain. It requires a macrodependency of a different form from the micro-dependencies that realize it, and its mark is cross-classification, where divisions at the macro level cut across divisions below.

That is a real constraint on the pressure case, and it excludes the simplest candidates. A macro-variable that is a relabeling of a single driver inherits that driver’s law and so fails novelty for the same reason the binary star does. What follows is therefore not the simplest proxy but the simplest one that survives their whole package.

Consider an upstream driver process with two components, U^1 and U^2 , independent of each other, each a symmetric two-state Markov chain on $\{0, 1\}$ that flips with probability q , where $0 < q < 1/2$. Two subsystems A and B track their respective components. The system’s specified dynamical ecology includes one control channel, a reset line on B with indicator R_t and value C_t . The update is simultaneous, and every variable at $t + 1$ is fixed by variables at $t + 1$ or t :

$$U_{t+1}^i = U_t^i \oplus N_{t+1}^i, \quad A_{t+1} = U_{t+1}^1, \quad B_{t+1} = \begin{cases} C_{t+1} & \text{if } R_{t+1} = 1, \\ U_{t+1}^2 & \text{if } R_{t+1} = 0, \end{cases}$$

where N^1, N^2 are independent Bernoulli(q) noise terms. The micro-state is $X_t = (U_t^1, U_t^2, A_t, B_t)$, and the reset line is an exogenous control input rather than a component of X_t : the controlled process is Markov on X_t once (R_t, C_t) is fixed. The

candidate macro-variable is the agreement of the two subsystems, $Z_t = A_t \oplus B_t$, taking the value 0 when they agree and 1 when they disagree.

The reset channel is part of the system's specification rather than an intervention imagined from outside it. In the observed process $R_t = 0$ for all t , so the subsystems are pinned to their components, $A_t = U_t^1$ and $B_t = U_t^2$, and $Z_t = U_t^1 \oplus U_t^2$. The restriction of the observed process to the synchronized manifold is therefore a fact about which channels happen to be exercised, not about what the dynamics permit.

Write $V_t = U_t^1 \oplus U_t^2$ for the driver parity and $r = q^2 + (1 - q)^2$ for the probability that the two noise terms agree. Substituting the update rule gives the identity the whole case turns on: with $R_{t+1} = 0$,

$$Z_{t+1} = V_t \oplus (N_{t+1}^1 \oplus N_{t+1}^2), \quad \text{so} \quad P(Z_{t+1} = V_t) = r.$$

The macro-variable's future is fixed by the *driver* parity, not by its own current value. The two coincide exactly when the channel is dormant, since $Z_t = V_t$ whenever $R_t = 0$. Everything below follows from that gap.

1. Unconditional relevance, taken with their indexing. Their condition ranges over the lower-level descriptions that subvene a given macrostate B and influence a macro-fact A , which together form an equivalence class (Franklin and Robertson 2024, §3.1). Fix A as $Z_{t+1} = 0$ and B as $Z_t = 0$. In the observed process the descriptions subvening B are exactly those with $V_t = 0$, namely $(0, 0)$ and $(1, 1)$, and each raises the probability of A from its unconditional value of $1/2$ to r , since $r - 1/2 = 2(q - 1/2)^2 > 0$ for $q \neq 1/2$. So $P(A \mid LLD) > P(A)$ holds across that class, which is what their condition asks. Descriptions with $V_t = 1$ lower the probability of A to $1 - r$, but they subvene $Z_t = 1$ and so belong to the class for a different macrostate.
2. Conditional irrelevance, in the strict rather than the approximate form. When $Z_t = 0$, the only micro-states the observed process ever visits are those with $V_t = 0$, and both yield $P(Z_{t+1} = 0) = r$. When $Z_t = 1$, the only visited realizations have $V_t = 1$, and both yield $1 - r$. Hence $P(Z_{t+1} \mid Z_t, X_t) = P(Z_{t+1} \mid Z_t)$ exactly. The proxy does not merely approximate screening off, in the way a deviant microstate would degrade it. It satisfies the strict condition.
3. Novelty, structurally rather than numerically. Z 's macrodependency relates parity classes to later parity classes, and neither its relata nor its transition partition coincide with those of either component process. The macrostate $Z = 0$ lumps $(0, 0)$ with $(1, 1)$, which cuts across the partition induced by either component's state, so the equivalence classes over which Z 's transitions are defined are not the classes over which U^1 's or U^2 's transitions are defined. This is cross-classification in Franklin and Robertson's sense, and it is why the case is not the binary star, whose centre of mass obeys the very dependencies its components obey. The numerical difference is a symptom of this rather than the argument for it: Z is a symmetric chain with persistence r against each component's $1 - q$, and $r - (1 - q) = q(2q - 1) \neq 0$ for $q \neq 1/2$. Were the parameters to coincide, the macrodependency would still not be either component's, because it would still be defined over different relata.

A defender may resist at the third condition, holding that parity is a mere algebraic consequence of two independent processes and so lacks the relevant kind of novelty. The reply is not that this reading is indefensible. It is that the reading is not the published one. If novelty requires more than cross-classification and non-identity of dependency profiles, whether organizational integration, explanatory unity, or something else, that further requirement has to be stated independently, because the probabilistic criterion as given does not supply it. A defender who adds it is conceding the paper’s point in the currency the paper asks for: the stated test underdetermines the verdict, and what settles it is a commitment the test does not contain. This is why the thesis is underdetermination rather than refutation.

So the stated implementation of the criterion returns a pass. That is weaker than saying the proxy belongs in the rainforest, and the weaker claim is the one the argument needs. What has been shown is that the conditions as stated, evaluated on the observed distribution, do not exclude this candidate. Whether Franklin and Robertson’s broader modal intentions would exclude it is exactly the question at issue, and the thesis here is underdetermination rather than refutation.

Now consider the micro-realizations of $Z_t = 0$ that the observed process never visits: those with $V_t = 1$. Take drivers at $U_t^1 = 0$ and $U_t^2 = 1$, and actuate the reset line at t with $R_t = 1$ and $C_t = 0$. Then $B_t = 0$, $A_t = U_t^1 = 0$, and $Z_t = 0$, while the drivers are untouched and $V_t = 1$. From there $P(Z_{t+1} = 0) = 1 - r$, against r for every observed realization of $Z_t = 0$, and $r \neq 1 - r$ for $q \neq 1/2$. Two micro-realizations of one macrostate carry different macro-futures, and the update rule shows why: the future tracks V_t , while the macrostate tracks V_t only for as long as the channel stays dormant.

Building the channel into the specification forecloses the natural objection, which is that the perturbation forces the system off the manifold on which the macro-variable was defined and so abandons the regime rather than testing it. It does no such thing. Actuating the reset line is an ordinary operation of the system’s own control ecology, the dynamics are unchanged throughout, the channel falls dormant again immediately, and the subsystems are repinned to their components at the next step, so the deviation is a one-step transient. The synchronized manifold was never a constraint the dynamics imposed. It was a restriction on the observed distribution produced by leaving an available channel unused, which is precisely the difference the criterion is designed to detect.

A harder version of the objection is worth meeting head on, since the countermodel invites it: any macro-variable would fail closure if one were free to posit enough internal reset channels. That is true, and it is why the admissibility constraints of §2.6 do the work they do. A channel cannot be introduced after the verdict is seen, must belong to the regime’s independently specified control ecology, must preserve the target regime, and must survive comparison with nearby defensible intervention classes. A reset line invented to embarrass a candidate fails the first and second of these immediately.

The objection is also a useful reminder of what the countermodel does and does not show. It establishes a possibility, that the stated observational criterion can certify a candidate whose autonomy is borrowed, and it locates the condition under which this happens: some admissible channel decouples the candidate from the structure

sustaining it. It shows nothing about how often that condition is met. Nothing here suggests that parasitism is widespread in the special sciences, and the argument does not need it to be. The conclusion is that the observational criterion is insufficient in principle wherever such channels exist, and whether they exist in a given domain is a question for that domain.

The proxy’s clean macro-law holds only because, in the observed process, $Z_t = 0$ guarantees that the drivers agree. The macro-variable never tracks that fact; it borrows it. Screening off quantifies over the realizations the process happens to visit, and on those realizations there is no heterogeneity to detect. Transition closure quantifies over the realizations an admissible intervention can reach, and there the heterogeneity is immediate. This is the same distinction the exact benchmark already draws: the partition is weakly lumpable, since the observed macro-process is genuinely Markov, but not strongly lumpable, since strong lumpability demands equal onward macro-transitions for every microstate in a class rather than only for those the observed distribution happens to weight.

The case is therefore not a verbal trick or a diagnostic failure. It is a countermodel: a candidate that satisfies unconditional relevance, strict conditional irrelevance, and novelty, and that nonetheless fails transition autonomy. Its role is to establish possibility, not frequency. Nothing here claims that such candidates are common in the special sciences, and the model is an illustration of a structural gap rather than evidence about any actual system. What it shows is that the gap is real: the stated screening-off test cannot by itself exclude borrowed autonomy, because the borrowing is invisible in the distribution the test quantifies over.

The natural rejoinder is that screening off, properly understood, already excludes the proxy. The reply to that rejoinder, developed in §2.2, is that existing screening-off diagnostics establish only the weak observational reading, while the proxy is excluded only on the strong interventional reading. The proxy meets the observational reading because the shared driver sustains conditional independence across time. It fails the interventional reading because an admissible perturbation targeting the proxy directly, with the drivers held intact, breaks the apparent autonomy and makes within-class micro-differences matter again for the ensuing macro-transition. Predictive or screening-off success in the proxy case depends on upstream organization that the candidate partition does not preserve under admissible perturbation, and that is what parasitism amounts to here.

3.3.2 A Second Pressure Case: Hidden-Feedback Parasitism

A second parasitism case arises when a macro-variable appears transition-autonomous because a fast hidden process continually relaxes it toward its apparent attractor. The source of the borrowed stability is now internal and fast rather than upstream and exogenous, but the skeleton of the previous case transfers directly, so the gap is not tied to one configuration.

Retain the two subsystems A and B and the candidate $Z_t = A_t \oplus B_t$, and replace the external drivers with two fast hidden variables H^1 and H^2 that relax toward the subsystems they track. The micro-state is $X_t = (A_t, B_t, H_t^1, H_t^2)$, and the dynamics are that each subsystem’s next value is its hidden variable’s current value, flipped

with probability q , after which the hidden variables re-equilibrate to the subsystems. Because the relaxation is fast relative to observation, the observed process never leaves the equilibrated manifold on which $H_t^1 = A_t$ and $H_t^2 = B_t$.

The verdicts are those of the previous case, and for the same reasons. On the equilibrated manifold each subsystem performs an independent symmetric walk with persistence $1 - q$, so Z is a symmetric chain with persistence $r = q^2 + (1 - q)^2$, which differs from $1 - q$ and cross-classifies the subsystem states exactly as before. Novelty is therefore satisfied on Franklin and Robertson’s condition rather than on the weaker reading that a dependence is stable at the macro grain, which is the reading their binary star already excludes. Within the observed process, each macrostate is realized only by equilibrated configurations, and those are homogeneous in their onward profile, so conditional irrelevance holds strictly. An admissible perturbation that acts faster than the relaxation sets a subsystem while its hidden variable is still stale. That reaches a realization of the same macrostate in which the hidden variables disagree, and from there $P(Z_{t+1} = 0) = 1 - r$ rather than r . Closure fails.

The perturbation is admissible in the same three respects: it acts on the candidate partition through a physically available channel, it leaves the dynamics untouched, and the system re-equilibrates within a step, so the regime is preserved and the deviation is a transient.

What differs between the two cases is the source of the borrowed stability. The latent-driver proxy borrows from an upstream driver that is exogenous and persistent; hidden-feedback parasitism borrows from an internal relaxation that is fast and endogenous. The shared skeleton is the substance of the claim. One structural fact generates the gap in both, that the observed process visits only a homogeneous subset of a heterogeneous macroclass, and the two sources show how little that fact depends on where the off-partition structure sits. A defender of the screening-off-plus-novelty package might concede the latent-driver proxy as an edge phenomenon while keeping the criterion intact. Two sources with the same signature are harder to relegate, because what is then at issue is not a peculiar configuration but a general category: apparent stability resting on structure the partition itself does not preserve under admissible perturbation.

The intervention test here is accordingly not just standard Woodwardian causal testing in new language. Standard interventionist analysis asks whether one variable causes another within an already accepted partition. The question here is prior to that: it asks whether a candidate partition itself carries autonomous transition structure under admissible perturbation.

Without an intervention-indexed test, genuinely autonomous macro-structure is indistinguishable from structure that merely inherits apparent autonomy from upstream. That is why admissibility must be declared before scoring rather than inferred from verdict patterns afterward, and it is the wedge motivating the rest of the paper.

Three consequences of the programmatic structured-space reading (§2.14) are worth flagging here, without making the core argument depend on them. Levels on this view are discovered features of system dynamics; they relate by nesting when closed grains stand in refinement relations; and the micro-level carries no privilege

from being smallest, since what matters for rainforest admission is causal autonomy under admissible perturbation, not microphysical size.

A similar point applies to causal-emergence work. Effective-information analyses identify cases where macro-descriptions can be more causally informative than micro-descriptions, often by improving effectiveness through changes in determinism and degeneracy (Hoel et al. 2013); Dewhurst (2021) argues that such gains support only epistemic emergence absent further constraints. That diagnosis is correct as far as it goes, but it identifies a gap rather than filling it. A partition can increase effective information while remaining fragile under intervention or horizon variation. The approach here treats such gains as corroborating evidence within a closure-governed criterion, not as a standalone ontology test.

Relatedly, Wallace’s dynamic treatment of higher-level ontology is right to push the debate away from mere collections of parts and toward structure “in the distributions and dynamics” (Wallace 2024). But a commuting or phase-space friendly representation is still not yet a full admittance criterion. Closure asks when the higher-level structure holds up under admissible perturbation rather than merely being calibrated to observed trajectories. Ladyman’s recent contribution to the real-patterns volume moves in the same dynamic direction. He characterizes real patterns in terms of “projectible generalisations that predict and explain” (Ladyman 2026, p. 1). The proposal here is that what remains to be added is explicit admission and downgrade discipline.

So the paper is not computational mechanics or causal-emergence analysis with new labels. It is a philosophical criterion that borrows formal tools while adding intervention-indexed commitment conditions, an explicit commitment protocol, and a sustained argument about why closure is the right fix for pattern-realism’s permissiveness problem.

4 Non-Ideal Application: Leakiness, Diagnostics, and Verdicts

4.1 Why Approximation is Expected

In complex systems, boundaries leak and couplings shift across regimes. Exact closure is therefore unusual. A credible realism criterion cannot require perfect closure everywhere. The study of approximate aggregation has a long history, especially Simon’s account of hierarchy and near-decomposability, where strong within-subsystem couplings can coexist with weaker cross-subsystem couplings and thereby separate short-run local dynamics from longer-run aggregate behavior (Simon 1962; Ando and Simon 1961). The extension here is philosophical rather than technical: it uses graded closure to support graded ontological verdicts, not to improve estimation. Rosas et al.’s distinction between informational, causal, and computational closure is useful here because it shows why the formal story is already graded and plural at the diagnostic level even when used to motivate one ontological criterion (Rosas et al. 2024).

Approximation is the expected non-ideal form of the same criterion, provided constraints are fixed and diagnostics are comparative.

Ontology tests should be type-level before token-level. A pattern type can be robustly closed for a regime even when a specific token fails because of boundary violation, atypical perturbation, or timescale mismatch. One anomalous token is therefore not decisive. The relevant question is whether failures are exceptional or systematic for the type under the stated constraints. Systematic failure, where within-class micro-differences repeatedly matter for macro-transitions across tokens under fixed constraints, is what triggers type-level downgrade rather than token-level exception.

4.2 Leakiness and the Distribution It Is Indexed To

Leakiness measures how much within-class micro-detail still improves macro-future prediction once the current macrostate is fixed. The natural quantity is the conditional mutual information $I(X_t; Z_{t+1} \mid Z_t)$ (Cover and Thomas 2006, eqs. (2.60)–(2.61)). In plain terms, a non-zero value means that residual micro-detail still helps predict the next macrostate even after the present one is fixed, which is the signature of a leaky partition.

Stated that way the quantity is incomplete, and the reason matters for the criterion rather than only for estimation practice. A conditional mutual information is defined relative to a joint distribution, and nothing in the notation says which distribution over within-class realizations is meant. Read against the observed joint, the measure inherits exactly the limitation the paper’s central argument turns on. The latent-driver proxy of §3.3 makes the point sharply: there, screening off holds strictly, so $I(X_t; Z_{t+1} \mid Z_t)$ is exactly zero under the observed joint, while the candidate fails transition autonomy. A measure that returns perfect closure for a candidate the criterion must reject cannot be the criterion’s target quantity.

The repair is to index leakiness to the distribution over within-class realizations, which yields two quantities with different evidential roles rather than one ambiguous one.

1. Observational leakiness: $I(X_t; Z_{t+1} \mid Z_t)$ evaluated under the observed joint. This is a screening diagnostic. It is cheap, it requires no interventional access, and a non-negligible value convicts immediately, since micro-detail that demonstrably improves macro-prediction within the observed process is transition-relevant heterogeneity the partition has failed to absorb. What it cannot do is acquit. A value of zero shows only that the realizations the process happens to visit are homogeneous, which is consistent with borrowed autonomy.
2. Admissible leakiness: the same functional evaluated under the distributions in which the within-class realization is set by an admissible intervention. This is the criterion’s quantity, and its domain has to be stated, since the verdict turns on it. For each macrostate z , let $R_I(z)$ be the microstates x with $g(x) = z$ that are reachable by interventions in the admissible class I while preserving the regime, and let $\mathcal{M}_I(z)$ be the finite-support distributions over $R_I(z)$ induced by randomized mixtures of admissible interventions. No single such distribution is privileged, so the quantity is a supremum over them and over macrostates, taken at the declared horizon L :

$$\Lambda_I^{(L)}(Z) = \sup_z \sup_{\mu \in \mathcal{M}_I(z)} I_\mu(X_t; Z_{t+1:t+L}),$$

where conditioning on Z_t drops out because μ is already confined to one macroclass, and the target is the L -step path rather than the next step alone, matching the criterion of §2.3. The supremum is what makes the quantity a test of the candidate: under an average, a partition could earn robust standing by being homogeneous on some convenient weighting of its realizations.

Two points about that definition need stating, since a reviewer will ask about both.

The kernels must be indexed to the interventions that produce them, not to a surgery on the whole micro-state. Writing the pairwise condition with $do(X_t = x)$ would misdescribe the admissible class and would sit badly with the restriction on Pearl-style surgery in §2.6: an admissible intervention here resets one subsystem and leaves the drivers untouched, so it never sets X_t wholesale, and not every member of $R_I(z)$ need be reachable by setting the micro-state directly. Define instead the post-intervention kernel

$$K_I(\cdot \mid x) = P(Z_{t+1:t+L} \in \cdot \mid X_t = x, \text{ admissible controls dormant thereafter}),$$

for $x \in R_I(z)$, and state closure as the requirement that $\sup_z \sup_{x, x' \in R_I(z)} D(K_I(\cdot \mid x), K_I(\cdot \mid x')) = 0$ for a divergence D separating distinct distributions.

The two formulations agree. If $\mathcal{M}_I(z)$ contains every binary mixture over $R_I(z)$, then $\Lambda_I^{(L)}(Z) = 0$ exactly when all realizations in $R_I(z)$ induce the same distribution over $Z_{t+1:t+L}$, for every z . One direction is immediate: if the kernels coincide on $R_I(z)$, then $Z_{t+1:t+L}$ is independent of X_t under any μ supported there, so the mutual information vanishes. For the other, suppose $K_I(\cdot \mid x) \neq K_I(\cdot \mid x')$ for some $x, x' \in R_I(z)$, and let μ put equal weight on the two. The conditional distribution of $Z_{t+1:t+L}$ then differs from its marginal on a set of positive measure, so $I_\mu(X_t; Z_{t+1:t+L}) > 0$ and the supremum is positive. Either form therefore vanishes just in case within-class micro-differences do not alter the macro-transition profile, which is transition closure as stated in §2.3. In the latent-driver proxy the observational quantity is exactly zero while both admissible forms are strictly positive, which is the separation the criterion turns on.

The two coincide whenever the observed process already visits the reachable realizations of each macrostate with appreciable weight, which is why observational diagnostics are so often adequate in practice and why the literature has had no occasion to separate them. They come apart exactly in the parasitism cases, where the observed distribution's support is confined to a homogeneous subset of a heterogeneous class.

This also puts the exact benchmark in its proper place rather than leaving it as a parallel formalism. Strong lumpability, in Kemeny and Snell's sense, requires that the coarse-grained process be Markov for *every* initial distribution, which is equivalent to requiring equal onward macro-transitions for every microstate in a class (Kemeny and Snell 1960). That is the limiting case in which admissible leakiness vanishes for every distribution supported on a macroclass. What observational leakiness can certify is weaker, and the contrast is best put without leaning on a term whose formulations vary with the initial distribution assumed: the observed process induces a Markov

macro-process for its actual initial or stationary distribution, while the partition fails strong lumpability over the full microstate space. The proxy is exactly such a case, and the two leakiness quantities separate on it for that reason.

The asymmetry between the two is the same one recorded in §2.11: predictive adequacy can persist where interventional stability fails, so predictive evidence is important but not final. Nothing here demotes observational leakiness to irrelevance. It remains the first thing to compute and the cheapest way to eliminate candidates. The claim is only that it screens rather than settles, and that a robust verdict requires the admissible quantity.

Using information-theoretic measures to state an ontological criterion might appear to assume a computationalist metaphysics. It does not. Both quantities are functionals of transition kernels, and the conditional independencies they measure hold or fail as facts about those kernels, whatever anyone computes and whether or not any system in the world exploits them (§2.12). No claim about the cost of tracking micro-detail, thermodynamic or otherwise, is needed to make the measure objective.

Admissible leakiness is the target quantity, and observational leakiness is its most accessible screen. Other diagnostics, such as within-class transition divergence and predictive gain from added micro-features, function as estimators or proxies when direct estimation is limited, and they inherit the same indexing: what they are evidence for depends on whether the within-class realizations they range over are those the process visits or those an admissible intervention can reach.

Leakiness should therefore be read comparatively and through robustness checks, not as a context-free absolute cutoff. Absolute tolerance is domain-relative, but high sensitivity of leakiness rankings to modest defensible modeling choices is itself a downgrade trigger.

Comparative ranking is necessary but not sufficient, since a candidate can beat nearby alternatives and still fail the criterion if absolute leakiness remains high and instability persists under modest robustness checks. Both relative superiority and minimum viability are required.

The viability floor can be stated without a numeric threshold. A partition fails minimum viability when, across the declared robustness checks, within-class micro-features yield consistent, non-negligible gains in macro-future prediction or intervention response relative to the best macro-only model. Persistence of that pattern across diagnostics and perturbation tests is decisive: the partition has not achieved the transition autonomy the criterion requires, regardless of how it ranks against competitors.

The term “non-negligible” can be given a comparative anchor without fixing a universal numeric threshold. A within-class gain is non-negligible when its magnitude is large enough that, were the same gain observed between two candidate macro-partitions, it would be recorded as a genuine comparative difference under the same diagnostic conventions. The viability floor therefore uses the within-class gain as a within-system comparandum rather than an external threshold: a partition fails if adding micro-features produces the kind of effect size that would itself count as diagnostic evidence against a rival candidate. Baseline model class and gain metric are

fixed upstream along with regime and admissibility under the canonical selection discipline (§4.3). The viability floor is applied first as a minimum gate; comparative ranking then assigns verdict categories among candidates that pass it.

Approximate closure is therefore not a discretionary replacement for exact closure. It is the non-ideal evidential route to the same ontological target. Judgment is constrained by upstream fixing of regime, admissibility, baseline model class, and gain metric, by comparative stability across modest robustness checks, and by explicit downgrade or indeterminacy when those rankings do not stabilize.

Verdict-category boundaries carry genuine vagueness because the underlying property of transition autonomy is itself graded. This is a feature rather than a defect: biological fitness is real and explanatorily significant without admitting a sharp boundary between fit and unfit. What matters primarily is ordering stability. When partition A consistently shows lower leakiness and greater perturbation stability than partition B across diagnostics, that ordering should be robust even if the exact placement of the boundary between “qualified” and “robust” carries some domain-conventional element. A partition that dominates alternatives across diagnostics and remains stable under modest changes to horizon, intervention distribution, and robustness checks earns the stronger verdict regardless of where one draws the categorical line.

4.3 Procedural Safeguards in Non-Ideal Cases

Approximate closure claims require explicit safeguards. The canonical selection discipline for the criterion, referenced throughout the paper, is as follows.

1. Fix regime, horizon, admissibility class, diagnostics, baseline model class, and gain metric before comparative scoring.
2. Compare candidate partitions under the same fixed setup.
3. Test robustness under modest changes in horizon and intervention distribution.
4. Apply a minimum-viability floor: if all candidates remain high-leak and fragile, return no robust standing verdict for that regime.
5. Report verdicts as robust, qualified, or indeterminate.
6. Disclose any post hoc revision to the setup and restart the evaluation rather than silently inheriting prior results.

These safeguards also fit Wimsatt’s robustness-analysis idea: confidence should rise when partially independent ways of detecting the same structure converge, and it should fall when apparent agreement is driven by shared assumptions or disappears under small perturbation (Wimsatt 1981). Multiple diagnostics do not replace criterion-level argument with a vote count. They reduce the risk that a favorable verdict is an artifact, or pseudorobust in Wimsatt’s sense, of one estimator, one horizon choice, or one admissibility specification.

This keeps threshold choice procedural rather than discretionary. It also addresses a predictable confusion. Estimation difficulty is an epistemic limit on access, not a defect in the metaphysical criterion itself. What is graded in non-ideal cases is not confidence alone, but the stability of closure itself across regime, horizon, admissibility, and robustness variation.

The purpose of this discipline is to make judgment auditable, not to eliminate it, and to prevent favored partitions from being protected by moving criteria.

4.4 Regime Dependence and Projectibility

Regime dependence is a claim about actual transition structure under specified constraints, not about observer-relativity. A pattern can be closed in one regime and leaky in another because the structure has changed, or because the available control channels have shifted. Since the admissible intervention class is defined by the regime's transition structure (§2.6), a change in that structure can alter both the admissible class and the closure verdict even when the system's micro-dynamics remain the same.

The metaphysical stance is explicit: regime and horizon index the modal profile being claimed, not a concession that anything goes.

An ordinary physical analogy helps. Ice crystals are objective structures only within certain temperature and pressure regimes. That regime dependence shows that objective structure can be conditional on equally objective background constraints without becoming epistemically subjective.

Projectibility provides a further check. Defensible regime specifications should support stable induction under modest counterfactual variation. Narrowly engineered regimes that protect a favored partition usually fail under small context shifts. When that occurs, the candidate was never robustly closed in the relevant sense.

Here the paper is closest to Ladyman and Ross. Projectibility is not treated as an optional methodological virtue. It functions as a realism-relevant stress test on whether a closure claim survives beyond calibration conditions (Ladyman and Ross 2007). Closure without projectible robustness is too cheap to support robust realist claims.

This is also why regime-indexing is not a concession to weakness. It is what allows robust verdicts in the special sciences to be disciplined rather than impossible. Without explicit regime and intervention indexing, most biological and social kinds would look merely leaky in an undifferentiated way. With that indexing in place, the question becomes whether the candidate is transition-autonomous within the bounded domain that gives it its explanatory role.

But projectibility is still not enough by itself, because a pattern can remain projectible while borrowing its stability from upstream structure (see §3.3). Closure asks the further question of whether the candidate partition carries its own transition structure under admissible perturbation.

Ladyman and Lorenzetti's effective ontic structural realism sharpens the same point at the metaphysical level. If ontology is regime-dependent and scale-relative, then the question is not whether a candidate belongs to a single final inventory of being, but whether it is stable enough in the relevant regime to merit commitment there (Ladyman and Lorenzetti 2024). Closure is offered here as the operational discipline for making that judgment explicit.

Regime dependence itself should be split into two forms. Ontic regime dependence occurs when system structure changes, such as phase transitions or boundary-condition shifts. Epistemic regime dependence occurs when measurement or control access changes while underlying structure remains fixed. The first changes what is

there to be tracked. The second changes what can be warranted from available evidence.

4.5 One Distributed Illustration

Macro does not mean large or spatially contiguous. Coarse-graining can be logical and distributed. Monetary systems illustrate this point without requiring new machinery. The cases below illustrate how the verdict categories operate; they are not worked empirical analyses of money or physiology, and the criterion's discriminating role does not turn on settling broader debates in either domain. A clarification is needed, given the earlier restriction to induced closure (§1.3). The relevant closure here is not the stipulated rules of the financial system (those are constitutive), but whether the physical and institutional infrastructure that implements those rules sustains autonomous macro-transitions under admissible perturbation. The failed-implementation case below shows exactly where induced closure breaks down even when the stipulated rules persist.

The macro-transition structure of transactions can remain stable across heterogeneous micro-realizations, such as cash tokens, ledger records, and digital balances, under admissible legal and financial interventions. If that stability holds, robust macro-level commitment is warranted in regime. If implementation channels degrade, closure can fail even while symbolic representations persist.

A minimal failure case clarifies the point. A state can retain formal legal tender rules while losing reliable payment enforcement and settlement implementation. Closure fails not because the rules changed but because the physical infrastructure that enforced screening-off is gone. The representation persists but transition autonomy degrades, and closure-based commitment should be downgraded. The same structure applies more broadly: the criterion explains when and why a social entity loses robust macro-level standing without requiring the claim that it was never real, or that it remains real simply because people believe in it. On the broader rainforest realist picture, the relevant issue is whether transactional organization still supports the economic structure in question, not whether the label survives institutionally ([Ladyman and Ross 2007](#)).

That is also the point of contact with the Ross-inflected side of rainforest realism about institutional kinds. Institutional ontology need not be arbitrary to be fragile. Ross's own point is narrower and game-theoretic: abstract strategic patterns can be real empirical objects without reduction to individual-level ontology ([Ross 1995](#)). What closure adds is a way of extending that thought to institutional kinds by asking which ones retain robust standing under heterogeneous realizations and admissible interventions, and which must be downgraded once enforcement, settlement, or trust structure no longer sustains the relevant macro-transitions.

The illustration does one job only. It shows that closure tracks transition structure, not spatial shape. Nothing in the argument depends on taking a stance on broader social ontology.

The contrast with the monetary case also fixes the shape of the three verdicts. Robust standing goes to candidates whose macro-transitions survive admissible intervention across heterogeneous realizers. Qualified standing goes to regime-fragile

candidates, such as organisms, whose screening structures hold across bounded horizons but degrade at pathological or ecological boundaries, which is why closure can be regime-robust without being regime-universal. Downgrade goes to candidates whose apparent stability is borrowed, such as the proxies of §3.3, or whose transition autonomy collapses once the supporting structure is perturbed, even where the representation persists.

4.6 Failure Conditions and Downgrade Rules

The account should also say clearly when commitment should be withdrawn or downgraded. Four failure patterns do that work.

1. Persistent high leakiness across defensible diagnostics under fixed constraints.
2. Strong disagreement among diagnostics that does not resolve under modest robustness checks.
3. High sensitivity of verdicts to small, defensible changes in admissibility or horizon.
4. Candidate sets where no partition clears the minimum-viability floor.

When these patterns appear, the right response is not forced binary judgment. The right response is explicit downgrade to qualified or indeterminate status. This keeps the criterion resilient without pretending that every case must produce a sharp ontology verdict.

Conversely, when closure persists across distinct, independently motivated admissibility classes within the same regime, that cross-admissibility stability raises evidential confidence for robust commitment.

Type-token reminder: a downgraded token does not automatically refute type-level closure. The key question is whether instability is exceptional at token level or systematic for the type under the stated constraints.

4.7 Stable Versus Merely Entailed Patterns

The graded account also supports an important distinction. A pattern can be entailed by microhistory and laws without being stable as a macro-handle. Entailment alone is cheap; stability under admissible perturbation is demanding. The distinction matters for permissiveness debates, because a contrived disjunctive construction can be true of a realized trajectory while still failing closure: it can require continual within-class micro repair, fail under modest regime shifts, and lose interventional reliability.

A historical case makes the pattern concrete. Phlogiston theory compressed combustion phenomena under a single macro-variable, but the partition required persistent bookkeeping to survive: when metals gained weight upon burning, theorists introduced “negative phlogiston”; when different substances showed different weight changes, further ad hoc parameters appeared. Each patch was a new within-class distinction imported to preserve macro-prediction. Phlogiston was, in the vocabulary of §3.1, a high-maintenance partition: reliable macro-prediction repeatedly forced refinement of within-class distinctions. By contrast, oxygen theory required substantially less such repair: the macro-variable (oxidation state) screened off the relevant

chemistry without persistent bookkeeping. The critique here is structural, not retrospective ridicule. A candidate that continually imports corrections to preserve macro-prediction behaves like a non-autonomous partition, and the closure framework detects this failure directly. The distinction between principled theoretical development and ad hoc rescue is itself contested philosophy of science, and the illustration is reconstructive rather than a clean empirical test. Closure supplies one principled way to draw the distinction, namely whether the supplementation requires persistent within-class micro-bookkeeping or introduces new transition-autonomous structure.

So gerrymandered constructions are not necessarily false; they usually fail to qualify for robust macro-level commitment. They may remain descriptions, but not ontologically weighty descriptions in the relevant regime. The same applies to computationally adequate models more broadly. A higher-level model can be useful for prediction or control without meeting closure standards. Computational adequacy means the model serves some purpose; closure requires that transition-relevant micro-differences are screened off in the specified regime. A computationally adequate but high-leak representation remains a valuable tool, but it does not automatically earn realist commitment. Closure is not a synonym for “good model.” It tests whether the relevant transition structure is in the world at the target grain, rather than merely encoded in a successful representation.

4.8 Practical Qualification Under Sparse Intervention Access

In many domains, direct interventions are sparse, ethically constrained, or expensive. Sparse intervention access changes evidential strategy, not the criterion itself. The evidential order is asymmetric: predictive success can raise confidence, but only interventional stability can underwrite robust commitment.

Where intervention data are limited, predictive closure functions as an operational indicator while interventional closure remains the ontological target. A model can fit historical trajectories and still fail under novel perturbation. For that reason, claims should be qualified by available intervention access and downgraded when out-of-regime behavior is unstable.

In-sample prediction is therefore not treated as decisive. The standard remains robustness under admissible perturbation, even when that robustness must be assessed indirectly.

Relevant evidence need not come only from humanly executable interventions. In observational sciences it can also come from naturally occurring disturbances, comparative cases, exogenous shocks, policy changes, and other perturbations that bear on the same transition structure. Such indirect intervention evidence often reveals whether a macro-description remains stable when the system is pushed off its ordinary trajectory.

This point also explains why diagnostics should be triangulated. Predictive closure can look strong in-sample while interventional closure fails under admissible perturbation. Partial observability can hide within-class heterogeneity that later appears as instability. Nonstationarity can make a previously low-leak partition unstable outside its calibration window. Triangulation does not remove these risks, but it makes them visible.

The monetary illustration is a useful case. A well-functioning payment system can look observationally stable for long stretches. What matters for stronger commitment is how that stability behaves under settlement delays, enforcement shocks, trust disruptions, or policy changes that function as admissible perturbations. Where those perturbations leave transactional organization intact, robust verdicts become plausible. Where the same structure survives only observationally, or collapses once such pressures appear, the right verdict is qualified rather than robust.

4.9 Conceptual Contrast: Near-Tie Prediction, Different Ontology Verdicts

Two partitions can show similar short-horizon observational performance and still receive different closure verdicts, and this is where the criterion does its real philosophical work.

Suppose partition P_1 and partition P_2 produce comparable one-step observational fit in calibration data. Under fixed admissible interventions, P_1 preserves stable macro-transition structure while P_2 requires repeated within-class micro refinements to maintain performance. Observationally, they may look close. Structurally, they are not.

On this account, P_1 receives the stronger commitment verdict because it remains transition-autonomous under the fixed constraints. P_2 can remain useful as a representation, but its dependence on recurring hidden repair counts against robust macro-level standing.

The latent-driver proxy of §3.3 is the exact instance of this contrast, and it is worth reading in that light. There the proxy and a transition-autonomous partition are not merely close on observational fit: the proxy satisfies strict conditional irrelevance, so no observational diagnostic separates them at all, and the verdicts diverge only once the admissible intervention class is applied. A near-tie in prediction is the general case; identity in prediction with divergence in closure is the limiting one.

The general pattern is familiar once stated. A partition calibrated to regularities that hold within a regime can match a transition-autonomous rival while the regime holds, and come apart from it under perturbations that redistribute those regularities without respecting the partition's boundaries. What separates the two is not predictive accuracy but whether the partition's own transition structure survives the perturbation or requires renewed within-class bookkeeping to be restored. That is the repair signature the criterion picks out, and it is why observational fit cannot settle admission.

4.10 Verdict Rule and Reporting Discipline

Verdict categories can be stated as a decision rule. A candidate earns:

1. *Robust standing* when it passes the minimum-viability floor (§4.2), survives modest horizon extension, remains stable across nearby defensible admissibility classes, and at least two diagnostics agree on its rank-order among candidates.

2. *Qualified standing* when closure evidence is positive but fragile across horizon or admissibility shifts, or interventional evidence is limited relative to the predictive leg.
3. *Indeterminate status* when diagnostics disagree persistently, rankings do not stabilize, or admissibility dependence is too strong for any determinate placement.

The rule is minimal rather than maximal: stricter domain conventions can narrow the robust category further, but nothing weaker than this triangulation should underwrite robust verdicts.

Where the canonical selection discipline (§4.3) fixes *what is held constant* before scoring, reporting fixes *what is disclosed* once scoring is done. To keep conclusions comparable across cases, closure assessments should state five items explicitly: target partition and rationale; regime and horizon specification; admissible intervention class and justification; diagnostics used and their agreement or disagreement; and final verdict category. This disclosure structure prevents silent shifts in target, timescale, or admissibility from being mistaken for genuine ontological progress.

The verdict vocabulary stays tied to ontology rather than to mere confidence. A robust verdict reports that a candidate occupies a stable place in the relevant closure structure under modest checks. A qualified verdict reports a fragile placement that can still support selective ontological commitment under explicit regime limitations. An indeterminate verdict reports that no stable placement has yet been secured. Rosas et al. provide the formal architecture for nested closure structure; the contribution here is to make that structure explicit in ontology reporting (Rosas et al. 2024).

Diagnostics do not replace criterion-level argument. They provide comparative evidence about whether a candidate satisfies the criterion under declared constraints. No single proxy is treated as infallible. Three checks are usually sufficient in practice: a leakiness proxy based on predictive gain from added within-class micro-features; within-class transition-profile divergence; and intervention-response invariance under admissible perturbation. Agreement across these checks increases warrant; persistent disagreement is evidence for revision or downgrade rather than for forced commitment. When interventional evidence is sparse or unavailable, the remaining diagnostics can still support qualified status, but robust standing requires that the interventional leg not be entirely missing.

5 Objections and Replies

The following objections move from foundational worries to evidential-discipline worries, but each receives the same core reply: closure under fixed constraints supplies ontological discipline without demanding a single privileged descriptive level.

Three standing commitments apply throughout the replies below and will not be restated each time. First, the account does not deny microphysical completeness; it claims non-redundancy at the explanatory and interventional level for declared macro-targets. Second, all verdicts are indexed to the declared constraints, so objections that presuppose context-free ontological claims address a view the paper does not hold. Third, the strongest conclusion is conditional on structural realist and interventionist

commitments; readers who withhold those commitments can still accept the narrower anti-gerrymandering result.

5.1 “Instrumentalism and Admissibility”

Objection: target selection is interest-relative, so the view still collapses into usefulness.

Reply: target selection can be interest-shaped without making closure verdicts preference-shaped. Once regime, horizon, and admissibility are fixed, whether macro-transitions are autonomous is a system fact under explicit constraints.

The stronger version invokes Dennett’s Martian (Dennett 1991, p. 42): if an ideal micro-predictor can forecast everything, macro realism looks optional. But this infers ontological redundancy from representational power. Even a micro-omniscient predictor coexists with objectively better macro-bottlenecks for target dynamics, and ignoring those bottlenecks increases representational burden without erasing macro transition structure.

Observer power changes convenience, not closure facts. Whether a partition is transition-autonomous under fixed constraints does not vary with who computes it.

Admissibility version of the same objection: admissibility criteria already encode what counts as the relevant level, so the circularity worry returns at one level up. The consolidated reply is that admissibility selection is not a free parameter. It is constrained by the system’s actual control ecology, which is to say the physically realizable, regime-preserving perturbations the system already faces under thermal fluctuation, diffusion, mechanical contact, and similar naturally occurring sources. The analyst’s task is to identify that ecology, not to construct it. Hierarchical order and disclosure rules make the identification auditable: admissibility is fixed from physical constraints before partition scoring, and post hoc admissibility revision triggers restart. Where the ecology is genuinely contested, the criterion correctly returns indeterminate verdicts rather than forcing a favored outcome. The worry that admissibility “shifts” the hard problem treats intervention-class choice as free; the framework treats it as a physically anchored feature of the regime being evaluated. Shifting is what happens when one problem is replaced by an equally hard one. Here the replacement is by a problem that is at least partly settled by the regime itself, since which control channels exist is not something the analyst decides. That does not make the choice automatic, and it is not claimed to.

Residual disagreement can remain. The account does not promise universal convergence from one pass. It promises disciplined comparison and explicit downgrade when verdicts are unstable across nearby defensible admissibility specifications. This is a feature for skeptical readers: hard cases are flagged as unresolved rather than settled by stipulation.

Instrumentalist pressure hides in two forms: ontology should track predictive utility only, or ontology is unnecessary once predictive utility is available. Both are wrong. Utility without closure is too permissive, and closure without ontology understates what stable intervention-guiding structure provides. What closure adds is commitment discipline, not metaphysical extravagance. If a candidate repeatedly supports stable counterfactual control under fixed constraints, withholding ontological standing is a

position that owes an account of what further condition realism was supposed to require. Rather than forcing maximal realism, the paper asks for explicit criteria for when anti-realism remains credible.

The strongest anti-instrumentalist point is structural rather than rhetorical. The objection that leakiness is “only” predictive error has real force against the observational quantity, which is why that quantity screens rather than settles (§4.2). Directed at admissible leakiness the objection presupposes a gap between tracking causal structure and describing what is real. But under the paper’s structural realist commitments, there is no further ontological layer for that gap to open onto. Stable counterfactual and predictive organization is not evidence for some deeper condition left unspecified elsewhere; it is what the relevant reality consists in at that grain. A partition that leaks under admissible intervention is not one where we failed to see the real object hiding underneath. It is one where the candidate lacks autonomous transition structure. Admissible leakiness therefore measures ontic deficiency in the candidate partition, not merely our ignorance. Once the test and the target are properly separated, the prediction miss is itself the structural fact.

The cheap-coding objection fails for the same reason. A recoding can improve local fit and still fail closure under admissible perturbation. The criterion is therefore representation-hostile in exactly the way permissiveness critiques demand.

5.2 “Causal Exclusion Still Defeats Macro Claims”

Objection: if microphysics is causally complete, macro-level causal claims are redundant (Kim 1998).

Reply: closure does not posit a second fundamental cause layer; it identifies when macro-description is sufficient for prediction and intervention at the relevant target level, leaving microphysical implementation untouched.

The standard reply to Kim’s exclusion argument is that macro-causation is explanatorily indispensable. The closure criterion allows a stronger, structural reply. Tracking every electron introduces non-difference-making noise into the causal model; the macro-boundary strips out this degeneracy and concentrates causal power at the level where difference-making actually occurs. The mechanism behind that reply is multiple realizability under closure: many distinct micro-configurations map to identical macro-outcomes, so the macro-variable is not a convenient shorthand competing with a fundamental micro-cause but a more informative causal parameter for the target outcome because it isolates the relevant difference-maker.

The earlier stages of Kim’s argument need to be engaged directly. Kim’s case rests on supervenience plus microphysical completeness: macrostates depend on microstates, and the micro-level is causally sufficient for its own effects, so macro-causation looks preempted. The closure reply grants both premises and denies that they yield the exclusion conclusion. Supervenience is indexed to instantaneous realization: every macrostate is fixed by some underlying microstate at a time. Closure-autonomy is indexed to an admissible perturbation class: under perturbations that respect the declared regime, within-class micro-detail adds no leverage for macro-transitions. These indices are different. A micro-level sufficiency claim about what produces a given effect token under the actual microstate does not undermine a

macro-level claim about which features are counterfactually stable across admissible perturbations, because the latter claim is about the class of micro-realizers that satisfy the macro-description, not about the particular realizer in play. Kim’s completeness premise secures bottom-up sufficiency; closure secures top-down non-redundancy under perturbation. Both can hold, and the exclusion conclusion follows only if one conflates token sufficiency with type-level redundancy. Once they are kept apart, supervenience grounds a dependency relation rather than an identity relation, and causal priority follows the level where conditional independence actually screens off within-class noise under admissible perturbation.

This reply also handles the fat-handedness worry that macro-interventions inevitably co-intervene on micro-states (Dewhurst 2021). Co-intervention on the micro-level is real, but closure ensures that the within-class micro-differences it produces do not change macro-transitions; the co-intervention is therefore non-difference-making for the macro target. Macro-level interventions can be successful and projectible across perturbations without denying microphysical realization. Woodward and Pearl supply the interventionist framework, while Hoel identifies a conditional route by which macro-variables can be more causally informative than their micro-descriptions (Woodward 2003; Pearl 2009; Hoel et al. 2013).

The overdetermination worry can be handled directly. The proposal does not claim two independent sufficient causes for one event. It claims one implemented process with multiple adequate descriptions at different explanatory and control targets. Exclusion pressure weakens once adequacy is indexed to target and intervention class rather than to a single privileged level.

Relative to Kim’s strongest formulations, the key move is target-indexing rather than layer multiplication (Kim 1998, 2005). A closed macro-description can be non-redundant without competing for micro-level fundamentality, because non-redundancy is inferred when micro-detail ceases to add control-relevant information for the target outcome under fixed constraints. The claim concerns which counterfactual dependencies remain stable for a declared class of interventions. Compatibility with microphysical completeness by itself is too weak to do this work, because it leaves open whether macro-variables are dispensable shorthand. Closure narrows that space: when it holds, dispensability must be argued against a stated record of interventional and predictive sufficiency, not simply asserted.

5.3 “Autonomy, Pluralism, and Distinctiveness”

Objection: the view sounds like a formal restatement of familiar special-sciences autonomy claims, not a distinctive realism thesis.

Reply: the paper is continuous with autonomy insights but not equivalent to them. Generic autonomy claims often remain permissive about what qualifies as a level-worthy grouping; this view adds a discriminating closure condition with explicit exclusion and downgrade rules, and that addition changes verdict structure. The difference is both practical and philosophical: practical because the account gives a protocol for ruling out parasitic and transition-incoherent candidates; philosophical because it ties realism commitment to transition autonomy under admissible interventions rather than explanatory convenience alone. In this respect, causal-emergence

style results are treated as evidence within a closure-governed approach, not as a replacement for the criterion itself (Hoel et al. 2013).

The same comparison can be made with the mechanisms literature. Mechanistic accounts are right to tie level talk to organized structure that supports productive organization and, in Craver’s interventionist development, stable intervention and explanation (Machamer et al. 2000; Craver 2007). Recent structuralist work on mechanisms makes the same pressure explicit: object-based metaphysics for mechanisms faces the problem of fitting special-science ontology to physics without collapsing it into mere convenience (Jiang 2024). But even that move does not by itself settle which variables or entities have been carved at the right grain for the target explanation. Closure addresses that earlier question by testing whether a candidate macro-variable screens off the micro-differences that would otherwise keep the mechanism explanatorily open.

The view should therefore be read as a constrained realism refinement of autonomy views, not as an unrelated alternative and not as a mere restatement. The conceptual difference can be stated directly. Generic autonomy claims assert that higher levels matter. Closure specifies *when* they matter, *how much* they matter (graded by leakiness), and *when the claim should be withdrawn* (explicit downgrade conditions). A realism criterion that specifies no withdrawal conditions is harder to assess, since nothing it rules out is stated in advance.

The classical arguments for the autonomy of the special sciences established that higher-level kinds can be explanatorily indispensable precisely because they abstract over heterogeneous micro-realizations (Putnam 1967; Fodor 1974). Closure entails multiple realizability as a structural feature: if a partition closes, distinct microstates within the same macroclass by construction produce identical macro-transitions, which is just what it means for the macroclass to be multiply realized in the dynamically relevant sense. Multiple realizability is therefore *necessary* for any closed macro-kind, but it is not *sufficient*: a purely extensional claim about kind membership says nothing about the dynamical coherence of the realizers. As Shapiro argues, not every appeal to heterogeneous realization earns explanatory autonomy (Shapiro 2000). A disjunctive aggregate can have heterogeneous realizers while completely failing closure, because those realizers can have utterly divergent transition profiles. What closure adds is the dynamical requirement: the realizers must not only fall under the same label but must produce the same macro-transitions under the declared constraints. This gap, between what a kind groups together extensionally and how those instances behave dynamically, is precisely where gerrymandered kinds exploit multiple-realizability intuitions without earning robust standing. The criterion formalizes a transition condition that helps separate genuine higher-level kinds from merely disjunctive aggregates (Shapiro 2000). The monetary illustration in §4.5 provides a concrete instance: coins, ledger entries, and digital balances are heterogeneous realizers, but what earns money robust macro-level standing is not the heterogeneity of realizers but the transition-equivalence of those realizers under admissible financial interventions.

Pluralism version of the objection: if multiple grains can satisfy closure, ontology becomes permissive again. The reply is that plurality of closed grains does not imply

arbitrariness. Once regime, horizon, and admissibility are fixed, which partitions close is determined by transition structure, not by analyst choice. When multiple candidates remain viable, robustness and minimality rank them as an objective relation among candidates under fixed constraints. In many domains these relations are further structured by coarse-graining order: when one closed partition is a refinement of another, their relation is nesting, not rivalry, which disciplines pluralism rather than dissolving it.

Andersen’s density-of-structure point helps explain why this is not an embarrassment (Andersen 2025). If, as she puts it, “the world may be dense with structure,” then one should expect more than one admissible closed grain in some systems (Andersen 2025, p. 1). The philosophical task is not to force uniqueness where the dynamics do not supply it. It is to report where in the closure space each candidate stands, how stable that standing is, and which grains are more robustly supported under the declared constraints.

This is a virtue for complex systems, not an embarrassment. Forcing a single grain in every context would be a stronger and less defensible claim than the paper needs.

5.4 “The Markov Template is Too Restrictive”

Objection: strong lumpability presupposes first-order Markov microdynamics and excludes memory-dependent systems.

Reply: the criterion is transition sufficiency, not first-order Markovity. Strong lumpability is an exact benchmark. In memory-bearing systems, state can be enriched with relevant history and the same closure question can be asked over that enriched state. This changes state representation, not criterion content, and the basic point has already been built into the exact benchmark discussion in Section 3.

A brief worked example. In adaptive immunity, a memoryless macrostate describing only current cytokine and lymphocyte concentrations fails closure: the same current state evolves very differently depending on whether a given antigen class has been encountered before, because primary and secondary responses follow different transition profiles. Enrich the state with one categorical history variable (prior-exposure status with respect to the relevant antigen class) and closure becomes informative at the target grain: the macro-future of the response is well-predicted by current state plus exposure category, and within-class micro-detail such as specific clonal identities or exact receptor sequences does not change macro-transitions across the bounded immunological regime. The history enrichment is real but small relative to the full microstate, so compression is preserved and the verdict is non-trivial. Institutional dynamics admit the same treatment, where path-dependent enforcement and trust trajectories supply the analogous history variable.

Minimality still matters. Enrichment should be the least extension needed to preserve closure performance under fixed constraints. The account also accepts an important edge case: if the minimally sufficient enriched state approaches micro-level complexity, then closure may be recovered without meaningful compression, which signals that no informative macro-partition has been found at the target grain rather than that robust standing has been achieved.

5.5 “Formal Closure Collapses the View into Formalism”

Objection: formal systems are closed by stipulation, so closure cannot ground empirical macro-level realism.

Reply: stipulated and induced closure must be separated. Formal systems can have objective internal closure under constitutive rules. The criterion here concerns induced closure in implemented spatiotemporal dynamics. Internal formal coherence does not by itself settle empirical macro-level commitment claims.

Formal systems are invented, but not freely. What counts as intelligible formal practice is constrained by stable identity conditions, compositional inference, and coherent consequence. Those constraints explain why formal work can guide empirical reasoning without itself deciding empirical standing.

No collapse follows. The paper’s argument is narrower and clearer: induced closure is the criterion of robust macro-level standing for implemented macro-patterns under declared regimes.

5.6 “Closure Is Too Strong and Eliminates Most Special Sciences”

Objection: if closure requires that within-class micro-differences not matter for macro-transitions, then most special-science kinds will fail. Biological kinds are notoriously leaky (Boyd 1991), and many psychological states and economic categories are no sharper. The criterion eliminates the ontology it was supposed to discipline.

Reply: the objection conflates robust standing with any positive verdict at all. Three verdict categories are available, not one. Robust standing requires stable closure under fixed constraints. Qualified standing fits cases where closure is partial, regime-limited, or supported by convergent but incomplete evidence. Indeterminate status fits cases where evidence is too unstable to warrant commitment.

Most special-science kinds fall into the qualified range, and that is the correct result. Biological species exhibit substantial transition autonomy within ecological and evolutionary regimes while leaking at boundary cases (hybrid zones, ring species, horizontal gene transfer). Psychological kinds often support predictive and interventional regularity within clinical or experimental regimes while failing under broader perturbation. The criterion does not eliminate these kinds. It gives them the status their closure evidence supports: qualified standing with explicit regime limitations, rather than either full robust status or wholesale elimination.

A common misunderstanding is that exact transition autonomy must hold everywhere if a special-science kind is to count as robustly real at all. That is too strong. Because the admissible intervention class helps define the regime, a macro-variable can achieve robust standing within a narrow but physically bounded ecological, physiological, or institutional home even if it leaks outside that home. A biological trait or economic variable need not be universally transition-autonomous to be robustly real. It need only be transition-autonomous under the intervention class that defines the regime in which it operates. Downgrade to qualified status becomes appropriate when those regime boundaries are stretched, or when intervention evidence within the regime remains too sparse to warrant the stronger verdict.

Nearby real-pattern work in biology provides a useful foil. One can characterize biological objects by their explanatory power and pragmatic utility, and by the availability of multiple valid carvings, without yet having shown that lower-level variation is transition-irrelevant in the stronger sense used here (Jiang 2025). Such utility matters, but it does not by itself settle robust standing.

The objection therefore proves too much. A criterion that granted robust standing to every special-science kind regardless of leakiness would be the permissive view closure is designed to replace. The gain from graded verdicts is that they match evidential reality rather than forcing a binary choice between full realism and eliminativism.

5.7 “Predictive Success and Failure Conditions”

Objection: a model can predict well without warranting ontological commitment.

Reply: that is correct, and the account does not deny it. Predictive success functions as evidence only when it aligns with closure diagnostics and interventional stability.

That point matters because otherwise the view would collapse into a redescription of successful modeling practice. It does not. A model can be descriptively successful, compressive, and operationally useful while still failing closure if its success depends on hidden heterogeneity, regime-specific repair, or borrowed upstream structure. Epistemic success is therefore not yet ontological standing.

This is also the natural place to distinguish the view from the strongest nearby tightening of real-pattern realism. Millhouse shows why compression, similarity, and counterfactual support already improve on a purely permissive reading of Dennett (Millhouse 2022). But even those stronger constraints leave open whether the candidate partition itself carries autonomous transition structure under admissible perturbation.

Two consequences follow. Diagnostics operationalize what the argument says should matter, but they do not replace criterion-level reasoning. And the criterion is falsifiable in its own terms: if candidate partitions repeatedly fail closure diagnostics across defensible regimes and admissibility classes, commitment should be withdrawn or downgraded. A proposal that cannot risk downgrade is not a serious realism criterion.

5.8 “Horizon-Relativity Makes the Criterion Too Weak”

Objection: if closure is indexed to horizon, any candidate can be made to pass at a short enough horizon.

Reply: horizon indexing is a constraint, not an escape clause. Two requirements prevent it from becoming one.

First, horizons must be anchored in independently identified timescale structure in the target domain: known relaxation windows, intervention latency, control-response periods, or similar physically motivated temporal features. A horizon chosen only because it makes a favored partition look closed has no independent standing.

Second, verdicts must survive a modest-extension stability check. A candidate that passes at horizon L but fails immediately at $L + \Delta$ for small, defensible Δ receives,

at best, qualified status. Robust standing requires that closure not be razor-thin in temporal scope.

A brief illustration makes this concrete. Suppose P_{lane} , the lane-segment partition of the commuter corridor of §2.7, is claimed at $L = 15$ minutes, a horizon chosen to cover typical control-response windows. The stability check runs the same diagnostics at $L = 20$ and $L = 30$ minutes, still within physically motivated timescales. Robust standing requires both that the comparative ranking against the rival P_{tod} not flip and that absolute leakiness for the leading candidate remain below the declared viability floor across the extended horizons. If P_{lane} dominates at $L = 15$ but the advantage vanishes at $L = 30$, the right verdict is qualified rather than robust: the closure holds within a bounded horizon but does not survive modest extension. If ranking and viability are preserved across L , $L + \Delta$, and $L + 2\Delta$, the check passes and the robust verdict is available.

Together, these requirements force explicit timescale discipline and prevent silent switching between incompatible temporal targets. If robustness fails, the verdict downgrades rather than the criterion breaking: the claim was too strong for the evidence.

5.9 “The Criterion Is Too Conservative”

Objection: by demanding closure and downgrade discipline, the criterion may be too conservative and may miss emerging macro-patterns.

Reply: conservatism is partly intentional. The paper aims to avoid premature ontological inflation. Still, the criterion is not rigid. It allows qualified verdicts for emerging patterns when closure evidence is partial but improving, and it allows verdict revision as regimes stabilize and intervention evidence accumulates.

The strongest version of the objection should also be granted. Some good macro-kinds enter science before robust intervention evidence is in hand. Early thermodynamic kinds are the obvious sort of case. The proposal should not be read as saying that such kinds were therefore unreal until a later evidential stage. The better verdict is often qualified: strong enough for selective commitment because the pattern is doing non-trivial explanatory work, but not yet strong enough for unqualified closure-based admission.

So the criterion does not require all-or-nothing maturity before any commitment. It requires that commitment strength track available closure evidence. Woodward (2021) similarly argues that while minimal interventionist criteria establish causality, invariance is a graded virtue. Our criterion of closure can be seen as the dynamical analogue of this: a macro-pattern is most defensible when it minimizes the omission of relevant micro-detail without collapsing the target structure into unnecessary fine grain. This is a virtue for dynamic systems where robust standing can be developmental rather than instantaneous.

Compression gains may therefore appear before robust closure is demonstrated; the account treats this as evidential lead, not ontological conclusion. Commitment can strengthen as closure evidence accumulates, and should remain qualified or be withdrawn when it does not.

6 Conclusion: A Disciplined Realism Claim

The three claims distinguished at the outset rest on different things. The anti-gerrymandering constraint rests on the wedge cases of §3.3 and requires no particular realism package. The criterion for robust commitment rests on the structural-realist and interventionist commitments of §1.2. The structured-space reading rests on neither, and is offered as a direction rather than a result: a reader who grants the first two can withhold it without loss to the admittance-and-downgrade protocol.

Rainforest realism is compelling, but its admittance criteria remain incomplete. Screening off plus novelty remains necessary, but observational success can be parasitic: the latent-driver proxy and hidden-feedback cases (§3.3, §3.3.2) show that a candidate can satisfy observational screening off while borrowing its apparent autonomy from off-partition organization the partition itself does not preserve under admissible perturbation. Closure under admissible intervention is what distinguishes autonomous from borrowed macro-structure. Strong lumpability provides the exact benchmark in Markov settings. Leakiness and convergent diagnostics extend the same criterion to non-ideal settings as a graded verdict protocol with explicit downgrade discipline.

This further step sharpens the non-redundancy demand already implicit in the Primacy of Physics Constraint (§2.5): a candidate is non-redundant when lower-level variation within its macroclasses stops mattering to the relevant macro-transition profile under declared constraints. In the recent literature, the criterion supplies what Franklin and Robertson’s screening-off test, Rosas et al.’s formal diagnostics, Ladyman and Lorenzetti’s effective realism, and recent dynamic formulations of pattern realism do not yet by themselves supply: an explicit closure-based admittance and downgrade discipline for robust macro-level commitment (Franklin and Robertson 2024; Rosas et al. 2024; Ladyman and Lorenzetti 2024; Wallace 2024; Ladyman 2026).

The proposal supports selective commitment: robust where closure is stable, qualified in borderline cases, explicitly downgraded where verdicts depend on fragile admissibility or horizon choices. The overall position is intentionally middle-range, stronger than compression-only pattern realism because it adds exclusion and downgrade structure, weaker than a total-level metaphysics because it stays restricted to induced closure in spatiotemporal systems under declared constraints.

Separating what is established from what is offered programmatically keeps the contribution accurately calibrated. The anti-gerrymandering role of closure, and the selector role that operationalizes the modal reading of screening off, stand on the argument of Sections 2 through 4. The further reading of rainforest realism as a structured space of computationally closed coarse-grainings is offered programmatically, in the spirit of the three-theses signpost (§1.3). Under that programmatic reading, the verdict categories make the structure reportable: robust where placement is stable, qualified where ontological support is real but fragile, and indeterminate where no stable placement has yet been secured. The formal richness of that space in any given domain depends on what the closure discipline turns up there, and the core argument does not depend on any particular such finding.

The view separates itself from eliminativist pressure by neither treating higher-level descriptions as fictions by default nor granting them standing by convenience

alone. The test is whether they track stable transition structure under admissible intervention. If they do, realist commitment follows on the paper's stated terms. If they do not, realism should be downgraded rather than protected by rhetoric.

This paper does not offer a complete metaphysics of levels: the scope stays restricted to induced closure in spatiotemporal systems under declared constraints. It does not settle general debates about emergence or reduction; closure is one necessary filter within a cumulative admittance protocol, not a standalone theory of higher-level reality. It does not supply a universal implementation recipe: applications beyond the illustrations depend on domain-specific estimator choices, robustness checks, and control-channel inventories that remain downstream of the philosophical argument and do not alter its content. And the strongest conclusion is conditional on the structural realist and interventionist commitments stated in §1; readers who withhold those commitments can still accept the narrower anti-gerrymandering result. Its contribution is a philosophical criterion with application discipline, not a panacea.

Future work should test the staged admittance protocol across domains by comparing candidate partitions under fixed constraints, rather than by multiplying new concepts. The paper has made its commitments explicit and addressed worries about circularity, instrumentalist drift, and scope ambiguity directly. If those replies fail, they fail on the merits of the stated argument rather than because those issues were left silent. One further question, left for future work, is whether this closure-based account of realist commitment bears on substrate-independence or on more abstract forms of structural standing.

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